

Introduction to Neural Scene Representation and Neural Rendering

Lingjie Liu



We Live in a World that is 3D and Contains Dynamics



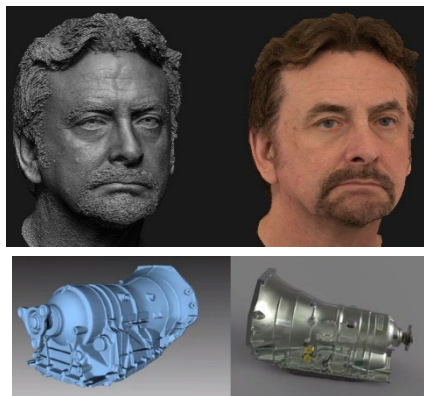
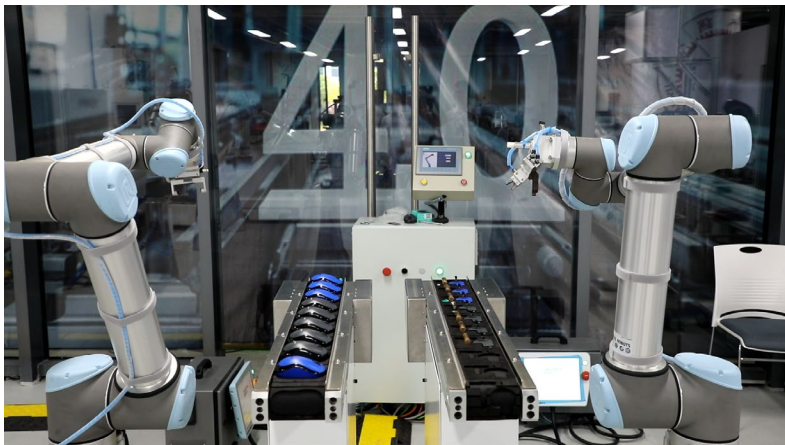
We Digitize Our World in 3D



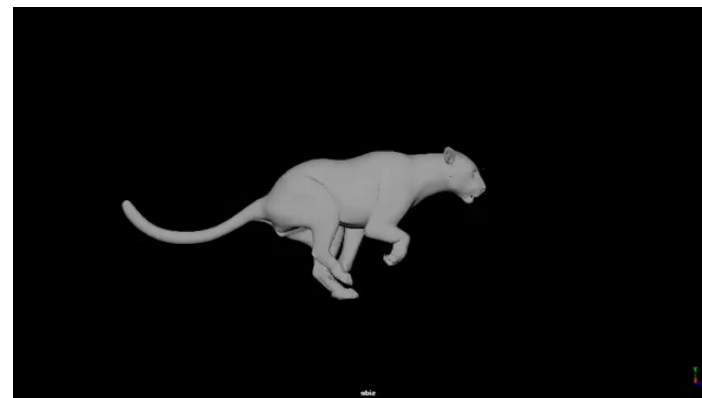
Future AI: Towards 3D Aware



3D Reconstruction of Real-world Scenes



Geometry
+ Appearance



Motion
+ Deformation

Photo-realistic Rendering

- **Image Synthesis** of Real-world Scenes with 3D Control.



Applications



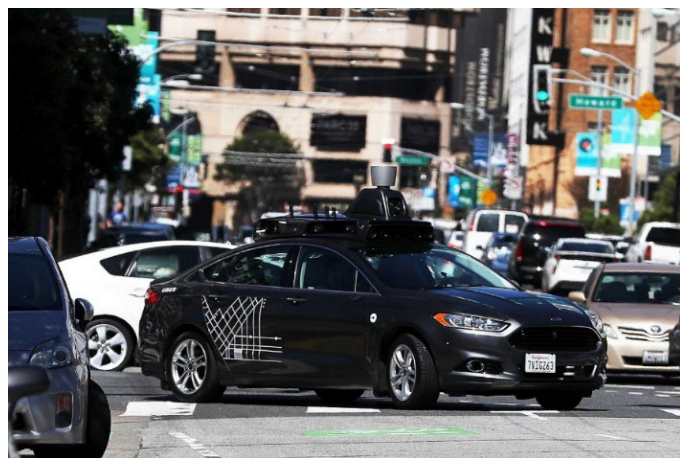
AR / VR



Gaming / Movie



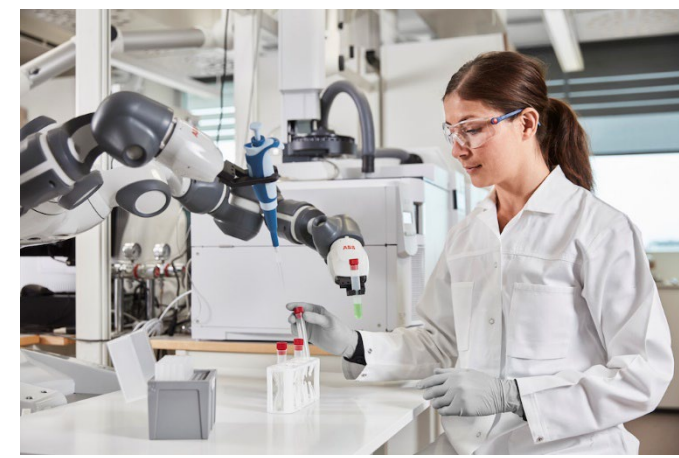
Healthcare



Autonomous Driving



Robot Grasping



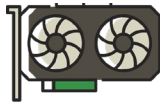
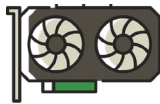
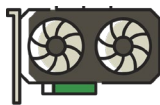
Human-robot Interaction

Why are they challenging?

Problem formulation

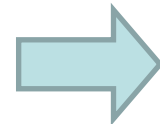
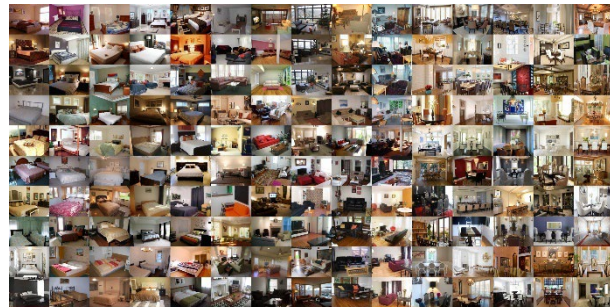


Captured images $\rightarrow$ Processing $\rightarrow$ Rendering of real-world place



[Mildenhall et al., Neural Radiance Fields (NeRF), ECCV 2020]
[Wu et al., Scalable Neural Indoor Scene Rendering, SIGGRAPH 2022]

Classical Computer Graphics Pipeline



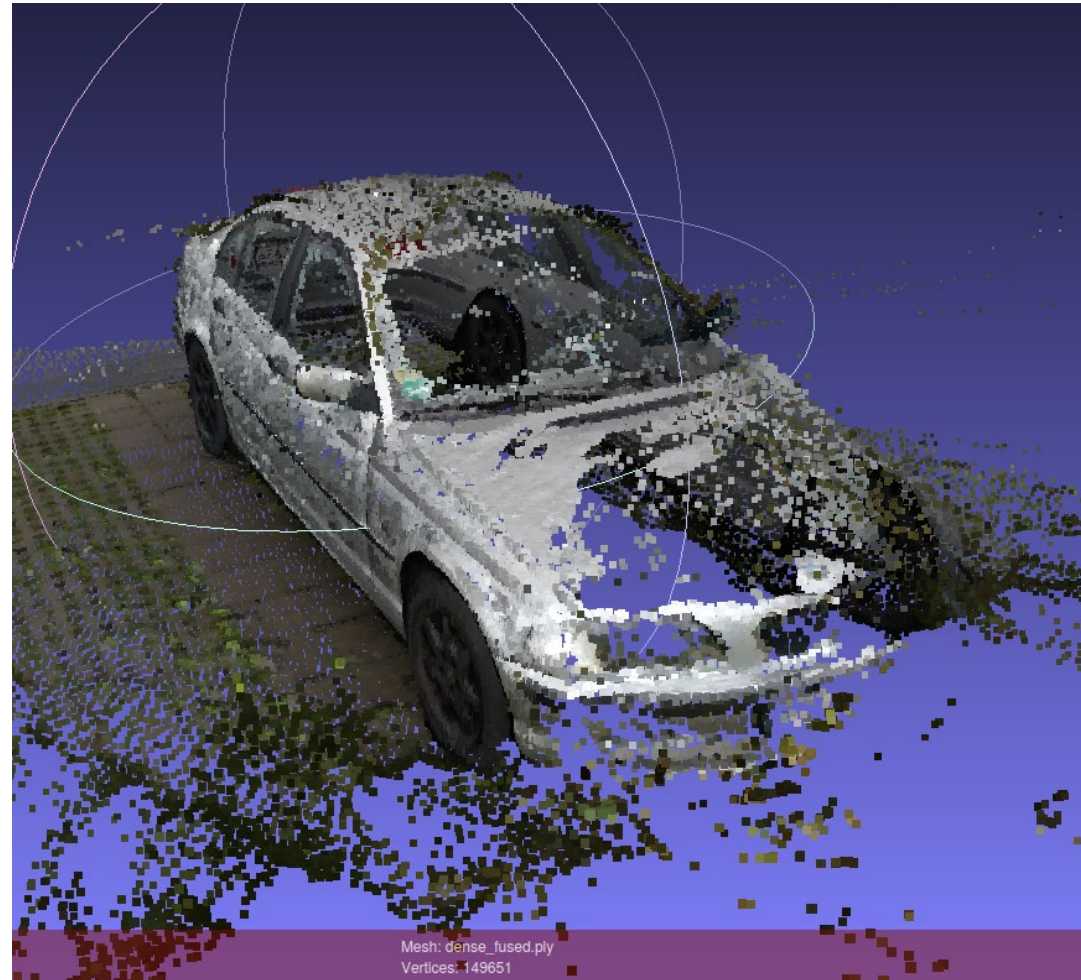
3D Reconstruction



Image Synthesis

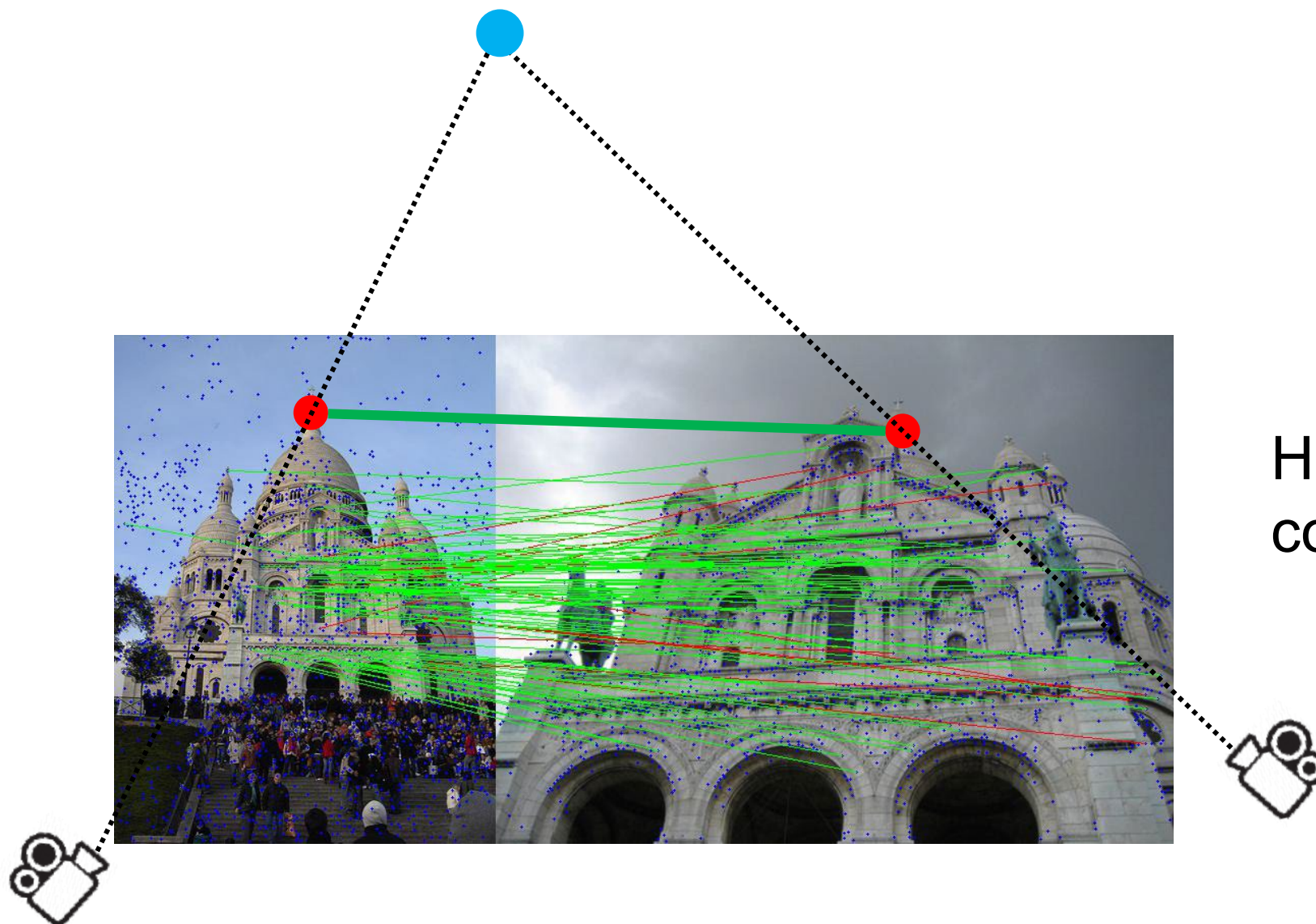
Computer Graphics Rendering

Image-based 3D Reconstruction



**COLMAP [Johannes et al. 2016, Schoenberger et al. 2016]
(Input: 100 images)**

Challenges in Image-based Reconstruction

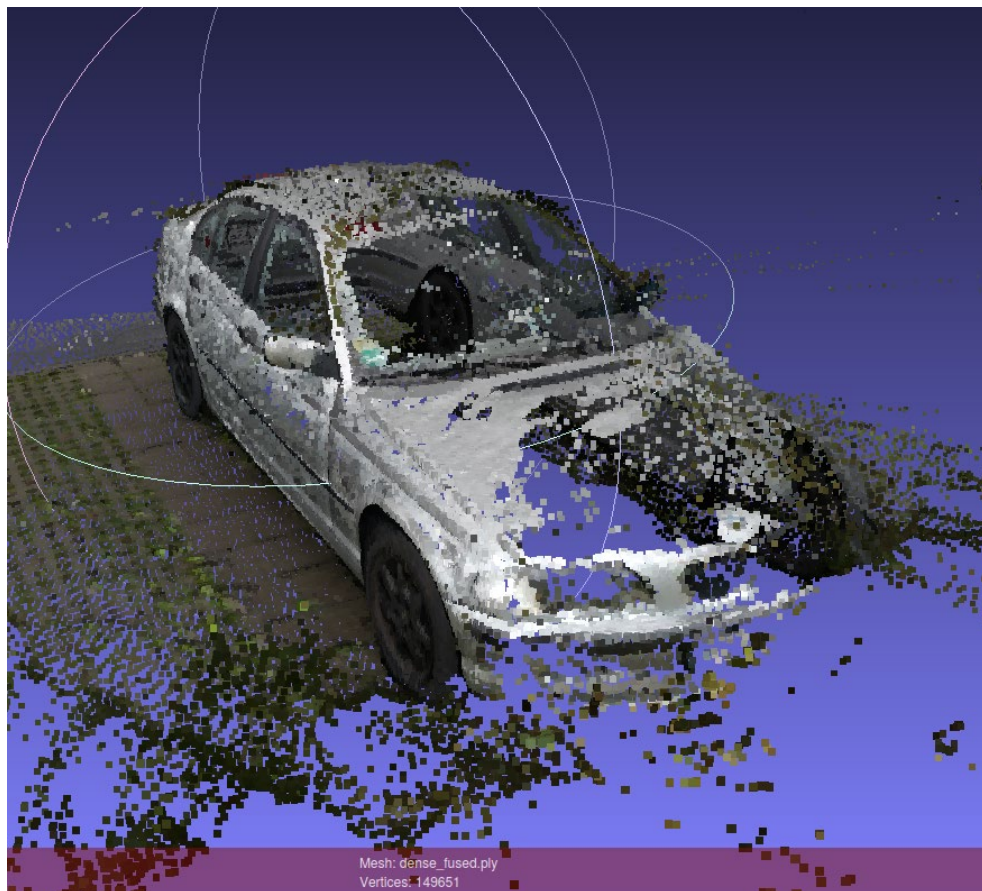


Hard to extract reliable correspondences!

Computer Graphics Rendering

Rendering requires very high-quality 3D models





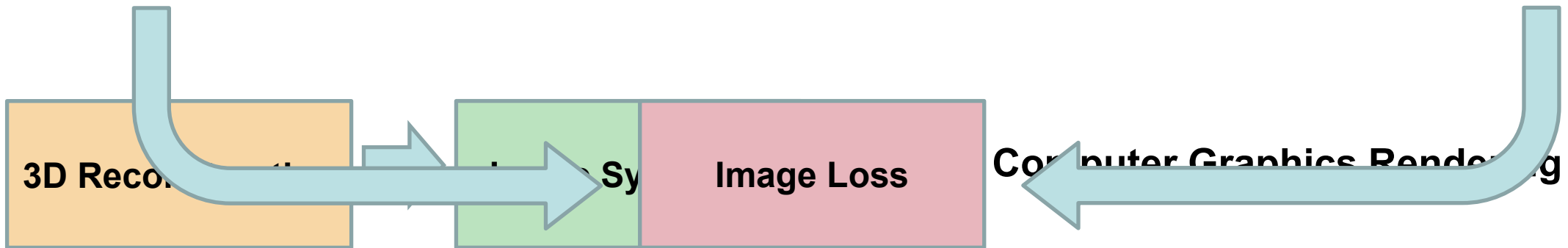
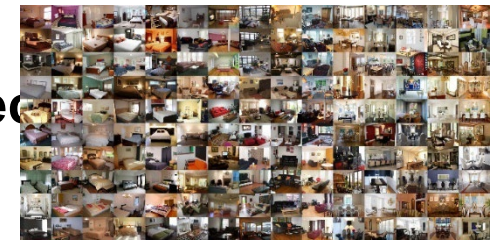
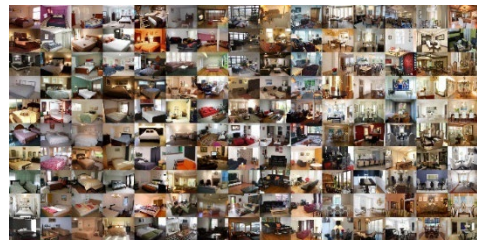
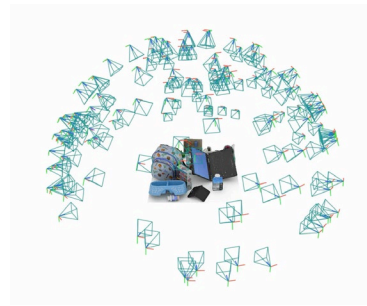
VS



Neural Scene Representation and Neural Rendering

To the rescue

Neural Scene Representation and Neural Rendering



Neural Scene Representation and Neural Rendering

DOI: 10.1111/egf.14022
EUROGRAPHICS 2020
R. Mantik and V. Sundstedt
(Guest Editors)

Volume 39 (2020), Number 2
STAR – State of The Art Report

State of the Art on Neural Rendering

A. Tewari^{1*}, O. Fried^{2*}, J. Thies^{3*}, V. Sitzmann^{2*}, S. Lombardi⁴, K. Sunkavalli⁵, R. Martin-Brualla⁶, T. Simon⁴, J. Saragih⁴, M. Nießner⁴,
R. Paidey⁶, S. Fanello⁶, G. Wetzstein⁷, J.-Y. Zhu⁸, C. Theobalt¹, M. Agrawala², E. Shechtman³, D. B. Goldman⁶, M. Zollhöfer⁴

¹MPI Informatics ²Stanford University ³Technical University of Munich ⁴Facebook Reality Labs ⁵Adobe Research ⁶Google Inc. ^{*}Equal contribution.

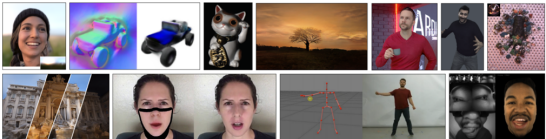


Figure 1: Neural renderings of a large variety of scenes. See Section 6 for more details on the various methods. Images from [SBT¹⁹, SZW¹⁹, XRS¹⁹, JRM¹⁷, GLD¹⁹, MBP¹⁹, LSMB¹⁹, MCK¹⁹, FTZ¹⁹, LIZ¹⁹, WSP¹⁹].

Abstract

Efficient rendering of photo-realistic virtual worlds is a long standing effort of computer graphics. Modern graphics techniques have succeeded in synthesizing photo-realistic images from hand-crafted scene representations. However, the automatic generation of shape, materials, lighting, and other aspects of scenes remains a challenging problem that, if solved, would make photo-realistic computer graphics more widely accessible. Concurrently, progress in computer vision and machine learning have given rise to a new approach to image synthesis and editing, namely deep generative models. Neural rendering is a new and rapidly emerging field that combines generative machine learning techniques with physical knowledge from computer graphics, e.g., by the integration of differentiable rendering into network training. With a plethora of applications in computer graphics and vision, neural rendering is poised to become a new area in the graphics community, yet no survey of this emerging field exists. This state-of-the-art report summarizes the recent trends and applications of neural rendering. We focus on approaches that combine classic computer graphics techniques with deep generative models to obtain controllable and photo-realistic outputs. Starting with an overview of the underlying computer graphics and machine learning concepts, we discuss critical aspects of neural rendering approaches. Specifically, our emphasis is on the type of control, i.e., how the control is provided, which parts of the pipeline are learned, explicit vs. implicit control, generalization, and stochastic vs. deterministic synthesis. The second half of this state-of-the-art report is focused on the many important use cases for the described algorithms such as novel view synthesis, semantic photo manipulation, facial and body reenactment, relighting, free-viewpoint video, and the creation of photo-realistic avatars for virtual and augmented reality telepresence. Finally, we conclude with a discussion of the social implications of such technology and investigate open research problems.

1. Introduction

The creation of photo-realistic imagery of virtual worlds has been one of the primary driving forces for the development of sophisticated computer graphics techniques. Computer graphics approaches span the range from real-time rendering, which enables the latest generation of computer games, to sophisticated global illumination simulation for the creation of photo-realistic digital humans in feature films. In both cases, one of the main bot-

tlenecks is content creation, i.e., that a vast amount of tedious and expensive manual work of skilled artists is required for the creation of the underlying scene representations in terms of surface geometry, appearance/material, light sources, and animations. Concurrently, powerful generative models have emerged in the computer vision and machine learning communities. The seminal work on Generative Adversarial Networks (GANs) by Goodfellow et al. [GPAM¹⁴] has evolved in recent years into

[Tewari et al. 2020]

This state of the art report is accepted at EUROGRAPHICS 2022.

Advances in Neural Rendering

A. Tewari^{1,6*}, J. Thies^{3*}, B. Mildenhall^{3*}, P. Srinivasan^{3*}, E. Tretschk⁴, W. Yifan^{4,8}, C. Lassner⁵, V. Sitzmann⁶, R. Martin-Brualla³,
S. Lombardi⁵, T. Simon⁴, C. Theobalt¹, M. Nießner⁴, J. T. Barron⁷, G. Wetzstein⁷, M. Zollhöfer⁴, V. Golyanik¹

¹MPI for Informatics ²MPI for Intelligent Systems ³Google Research ⁴ETH Zürich ⁵Reality Labs Research ⁶MIT ⁷Technical University of Munich ⁸Stanford University ^{*}Equal contribution.



Figure 1: This state-of-the-art report discusses a large variety of neural rendering methods which enable applications such as novel-view synthesis of static and dynamic scenes, generative modeling of objects, and scene relighting. See Section 4 for more details on the various methods. Images adapted from [MST²⁰, TY20, CMK²¹, ZSD²¹, BBF²¹, LSS²¹, PSB²¹, JXX²¹, PDW²¹] ©2021 IEEE.

Abstract

Synthesizing photo-realistic images and videos is at the heart of computer graphics and has been the focus of decades of research. Traditionally, synthetic images of a scene are generated using rendering algorithms such as rasterization or ray tracing, which take specifically defined representations of geometry and material properties as input. Collectively, these inputs define the actual scene and what is rendered, and are referred to as the scene representation (where a scene consists of one or more objects). Example scene representations are triangle meshes with accompanied textures (e.g., created by an artist), point clouds (e.g., from a depth sensor), volumetric grids (e.g., from a CT scan), or implicit surface functions (e.g., truncated signed distance fields). The reconstruction of such a scene representation from observations using differentiable rendering losses is known as inverse graphics or inverse rendering. Neural rendering is closely related, and combines ideas from classical computer graphics and machine learning to create algorithms for synthesizing images from real-world observations. Neural rendering is a leap forward towards the goal of synthesizing photo-realistic image and video content. In recent years, we have seen immense progress in this field through hundreds of publications that show different ways to inject learnable components into the rendering pipeline. This state-of-the-art report on advances in neural rendering focuses on methods that combine classical rendering principles with learned 3D scene representations, often now referred to as neural scene representations. A key advantage of these methods is that they are 3D-consistent by design, enabling applications such as novel viewpoint synthesis of a captured scene. In addition to methods that handle static scenes, we cover neural scene representations for modeling non-rigidly deforming objects and scene editing and composition. While most of these approaches are scene-specific, we also discuss techniques that generalize across object classes and can be used for generative tasks. In addition to reviewing these state-of-the-art methods, we provide an overview of fundamental concepts and definitions used in the current literature. We conclude with a discussion on open challenges and social implications.

1. Introduction

Synthesis of controllable and photo-realistic images and videos is one of the fundamental goals of computer graphics. During the last decades, methods and representations have been developed

to mimic the image formation model of real cameras, including the handling of complex materials and global illumination. These methods are based on the laws of physics and simulate the light transport from light sources to the virtual camera for synthesis. To this end, all physical parameters of the scene have to be known for

arXiv:2111.05849v2 [cs.GR] 30 Mar 2022

[Tewari et al. 2021]

Neural Rendering - Definition

- Definition:

*"Deep neural networks for **image or video generation** that enable **explicit or implicit control** of **scene properties**"*

1)

Generative networks that
synthesize raw pixel output

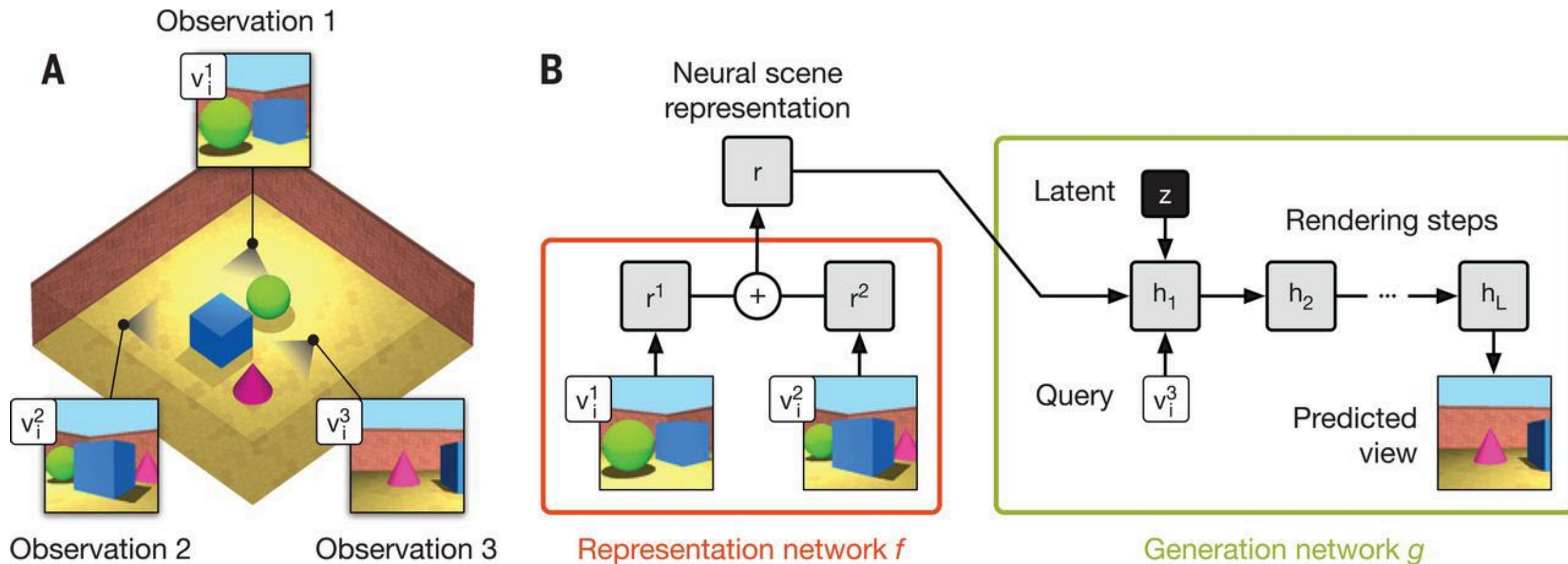
2)

Controllable by
interpretable parameters
or by video/audio input.

3)

Illumination, camera, pose,
geometry, appearance, or
semantic structure controllable

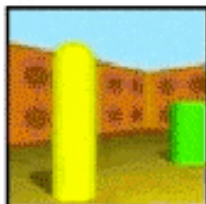
Generative Query Network (GQN)



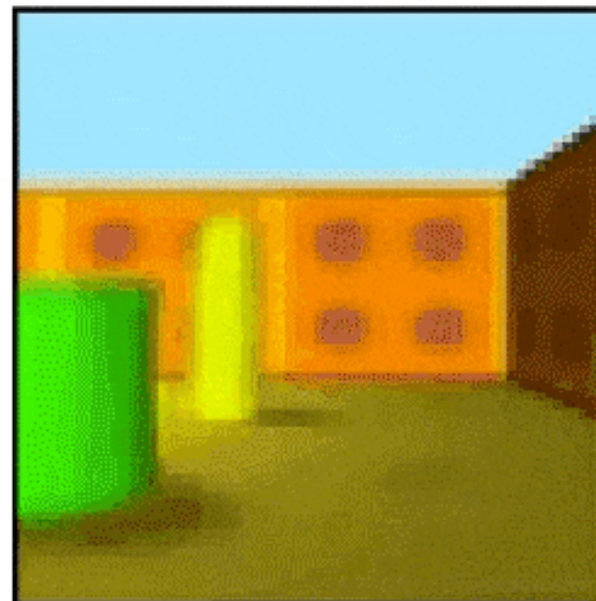
Neural scene representation and rendering, Eslami et al. 2018

Generative Query Network (GQN)

observation



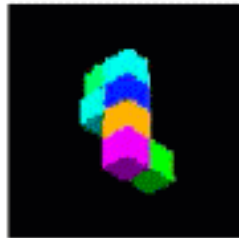
neural rendering



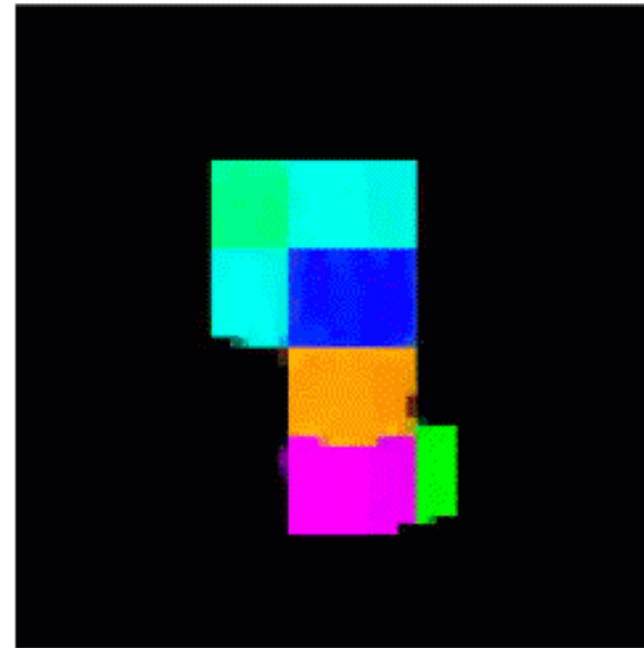
Neural scene representation and rendering, Eslami et al. 2018

Generative Query Network (GQN)

observation

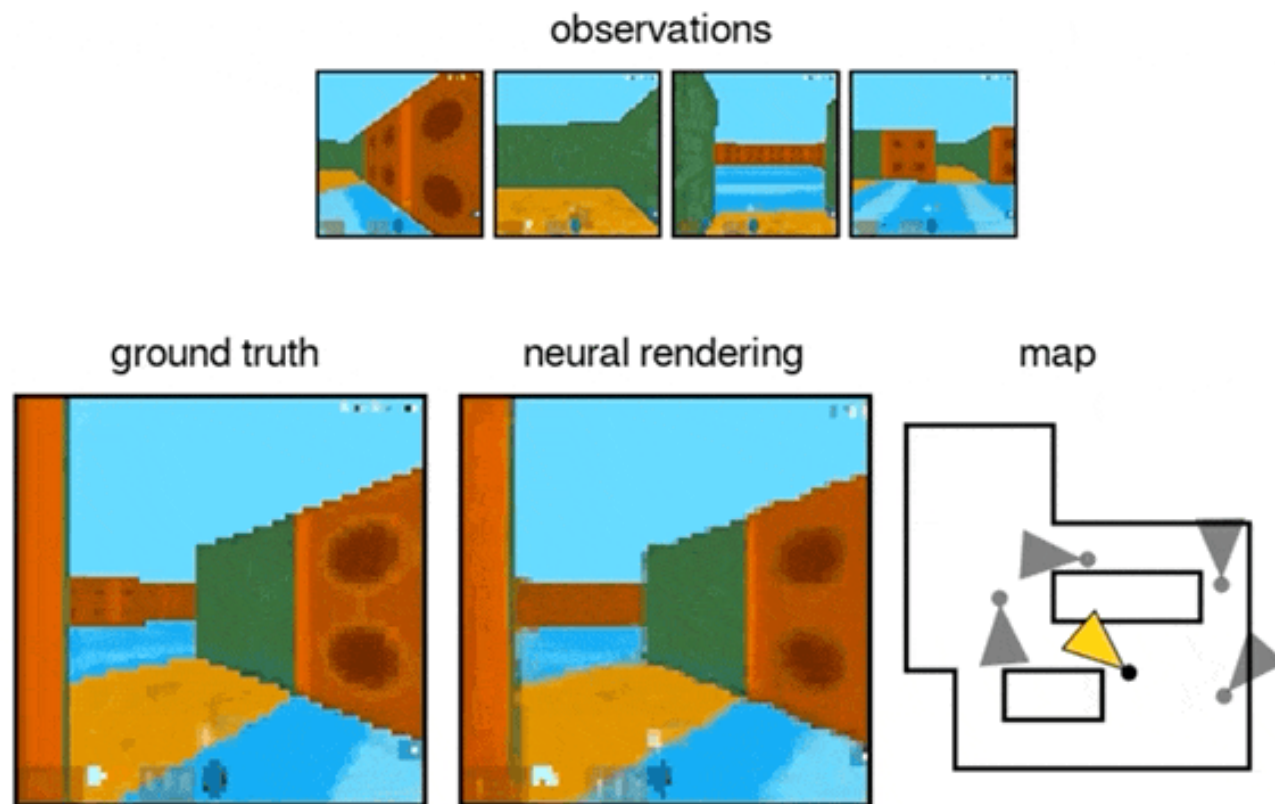


neural rendering



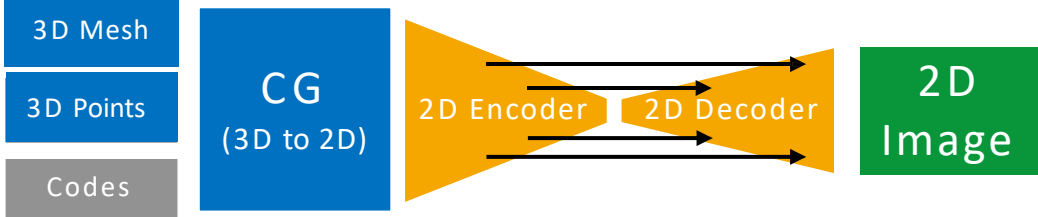
Neural scene representation and rendering, Eslami et al. 2018

Generative Query Network (GQN)

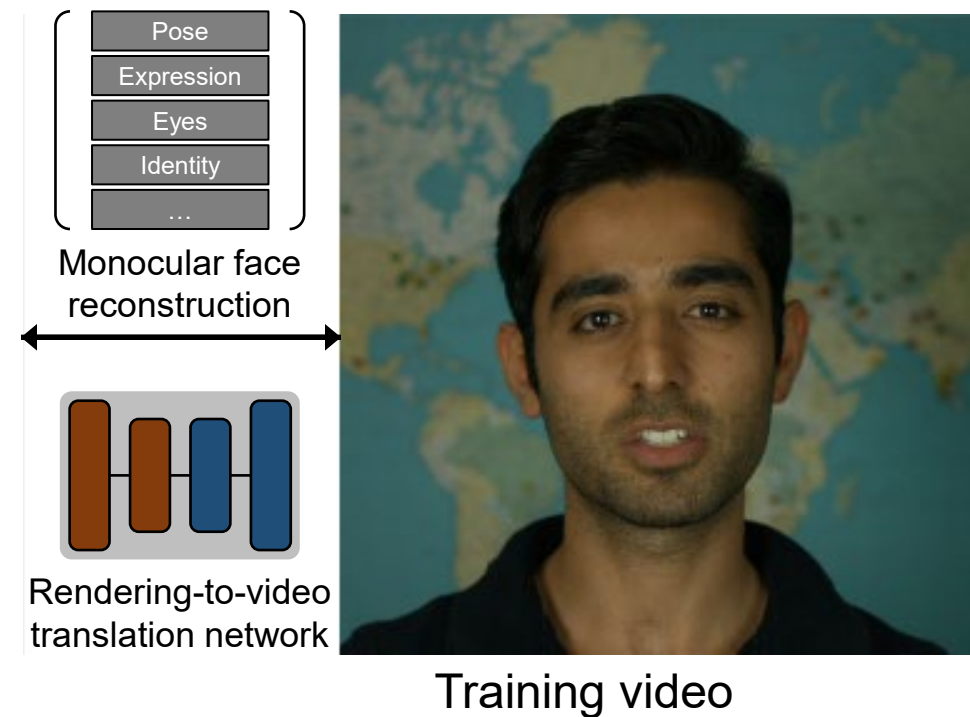


Neural scene representation and rendering, Eslami et al. 2018

Neural Rendering Zoo

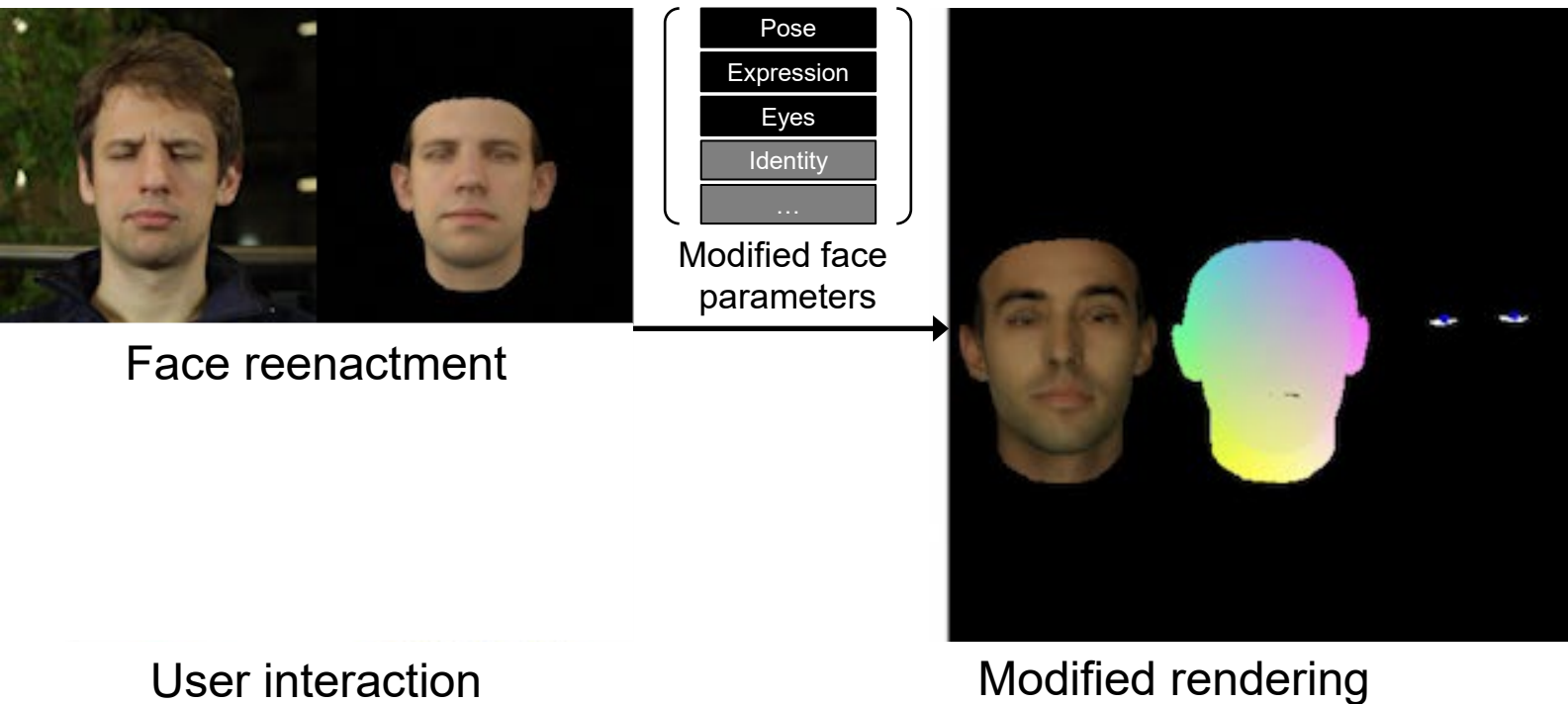
“Regress it”		e.g., GQN
“Make it more real”		e.g., DVP or DNR

Deep Video Portraits (DVP)



Deep Video Portraits, Kim et al. 2018

Deep Video Portraits (DVP)



Deep Video Portraits, Kim et al. 2018

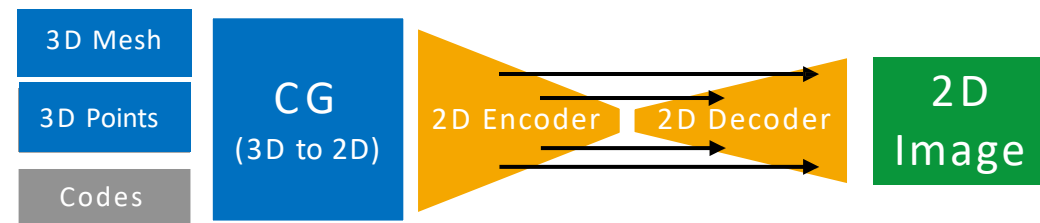
Neural Rendering Zoo

“Regress it”



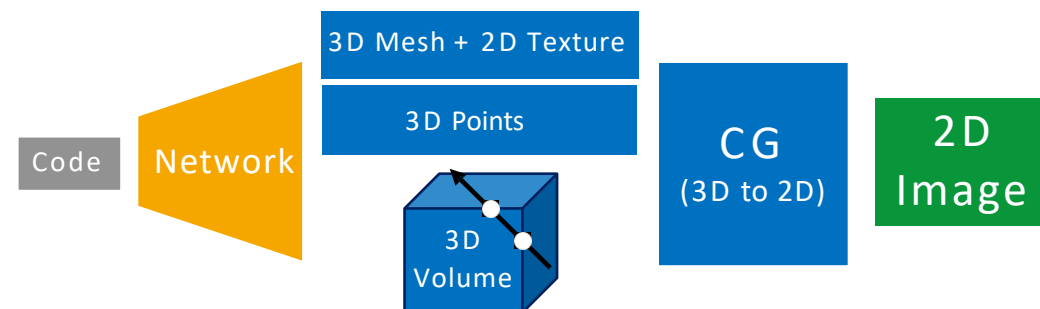
e.g., GQN

“Make it more real”



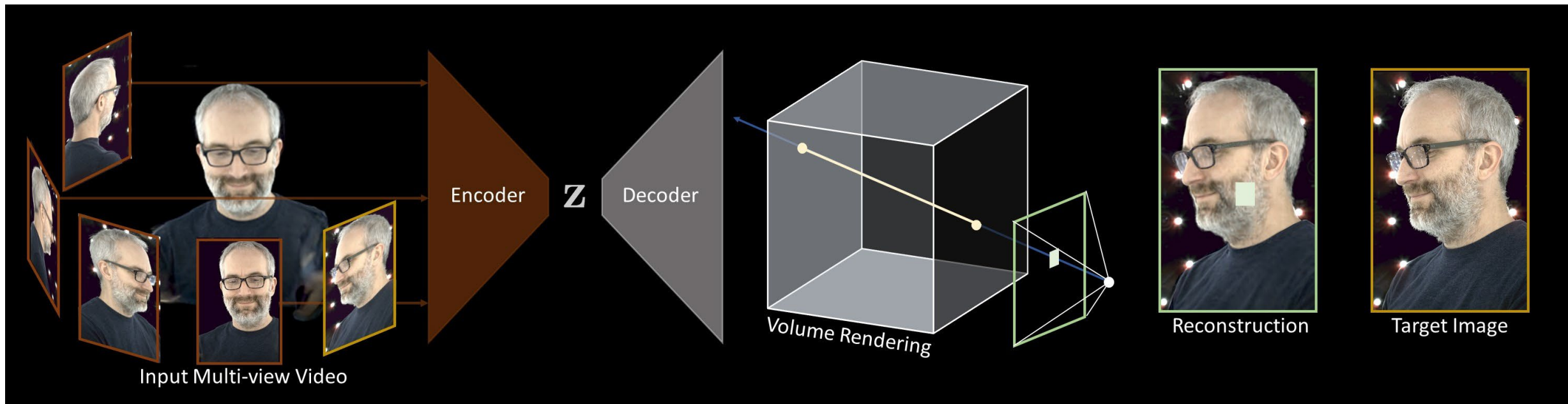
e.g., DVP or DNR

“Regress & render”



e.g., Neural Volumes

Neural Volumes

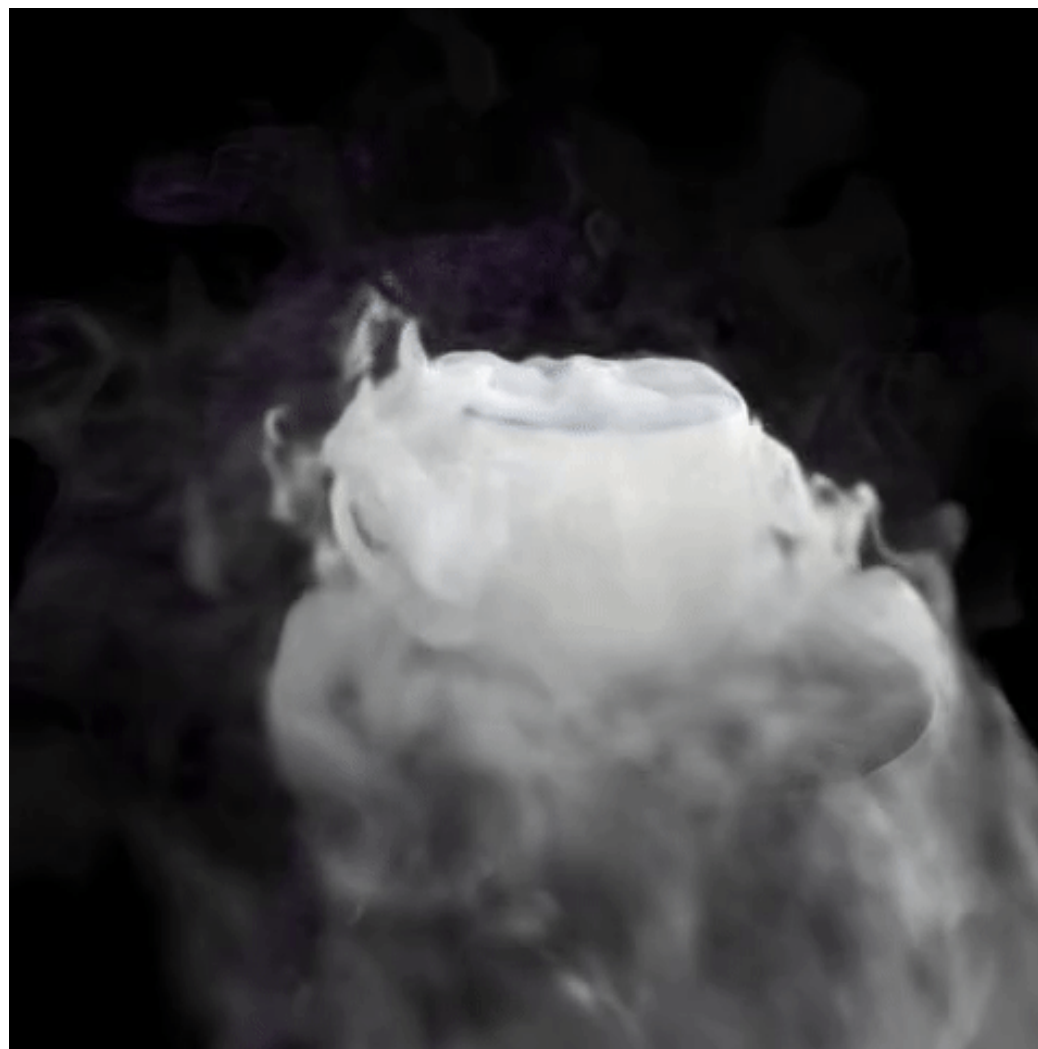


Neural Volumes: Learning Dynamic Renderable Volumes from Images, Lombardi et al. 2019

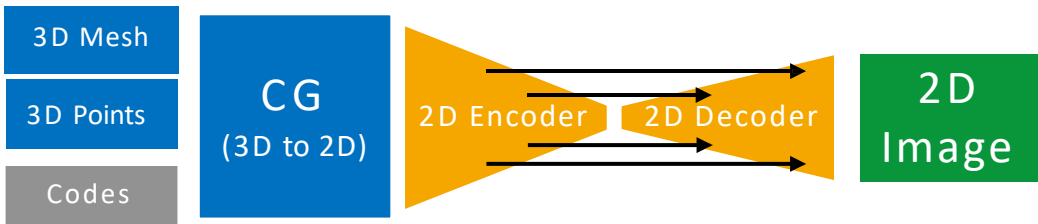
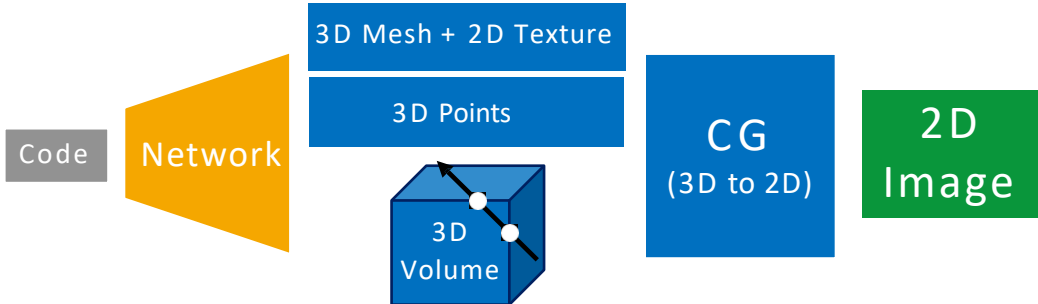
Neural Volumes



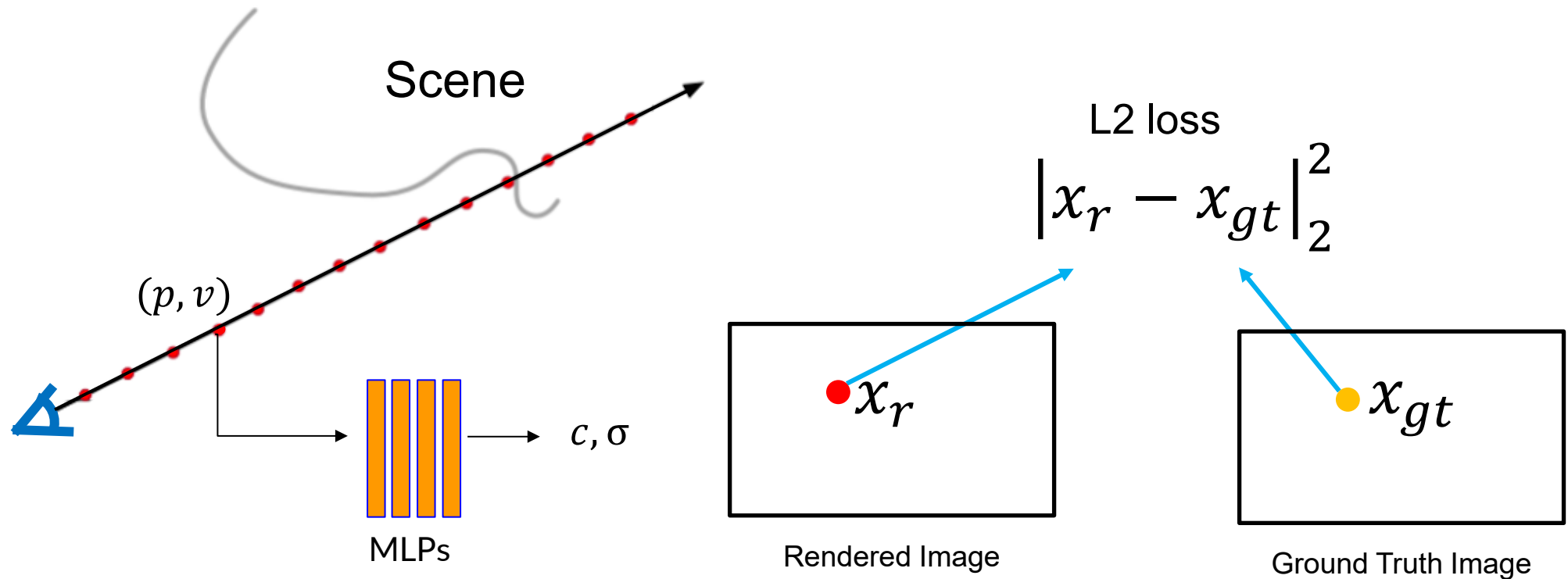
Neural Volumes



Neural Rendering Zoo

"Regress it"		e.g., GQN
"Make it more real"		e.g., DVP or DNR
"Regress & render"		e.g., Neural Volumes
"Step, sample & blend"		e.g., NeRF

Neural Radiance Fields (NeRF)



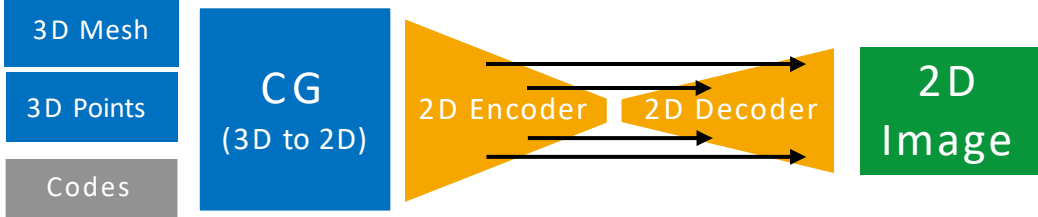
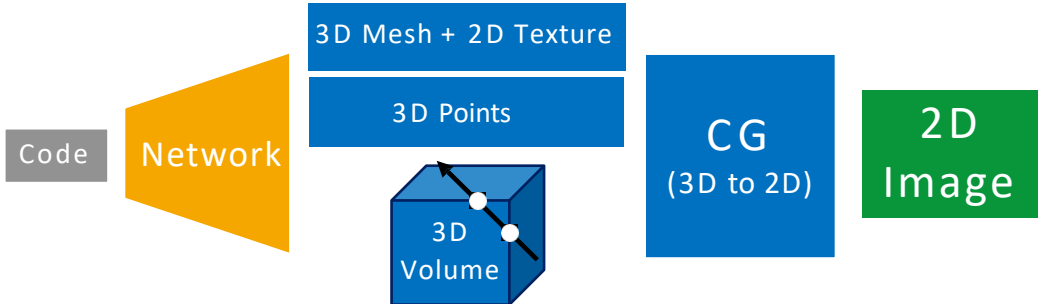
[Mildenhall et al. 2020]

Neural Radiance Fields (NeRF)

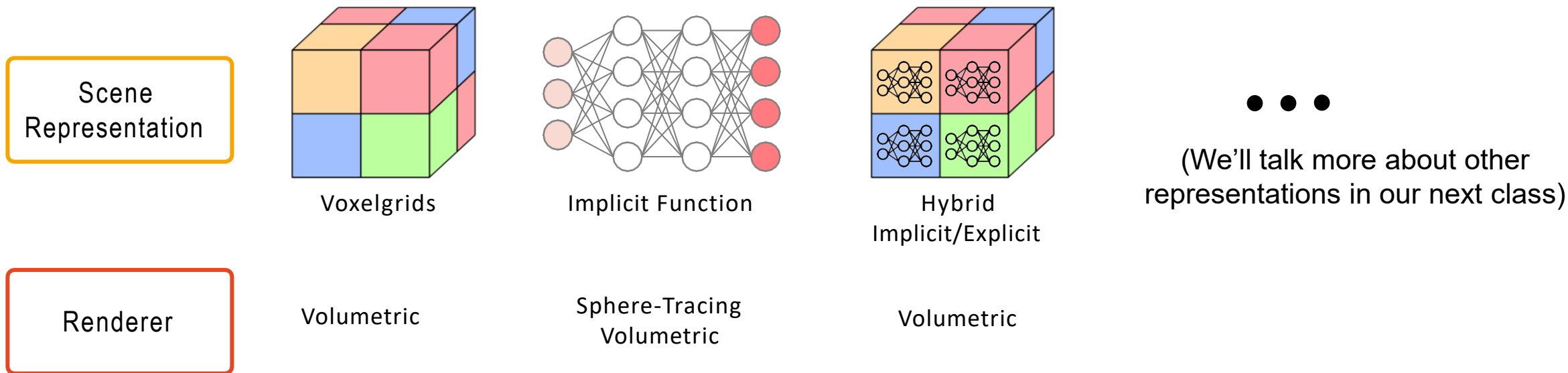


[Mildenhall et al. 2020]

Neural Rendering Zoo

“Regress it”		e.g., GQN
“Make it more real”		e.g., DVP or DNR
“Regress & render”		e.g., Neural Volumes
“Step, sample & blend”		e.g., NeRF

Overview

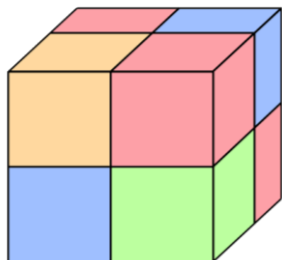


Both Scene Representation and Differentiable Renderer often adapted from traditional computer graphics.

Requirements

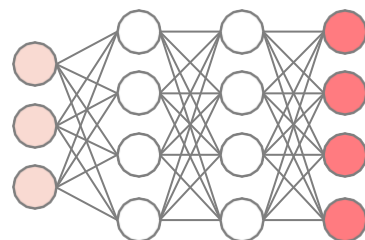
Scene
Representation

Renderer



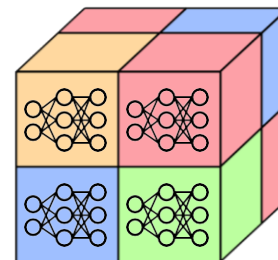
Voxelgrids

Volumetric



Implicit Function

Sphere-Tracing
Volumetric



Hybrid
Implicit/Explicit

Volumetric

...

Pros

Cons

Voxel-based methods

DeepVoxels



Sitzmann et al., CVPR 2018

Neural Volumes



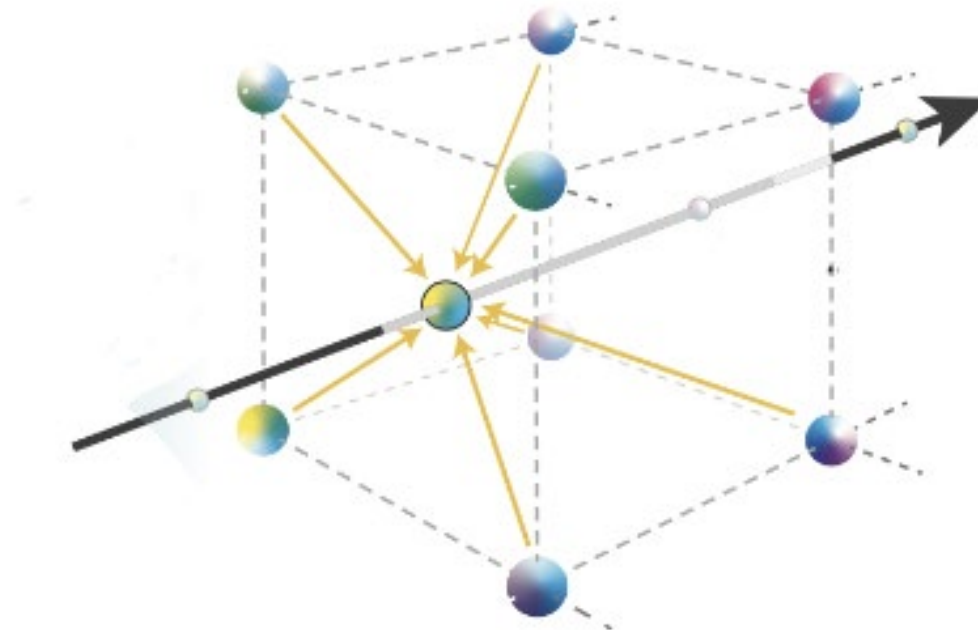
Lombardi et al., SIGGRAPH 2019

HoloGAN



Phuoc et al., ICCV 2019

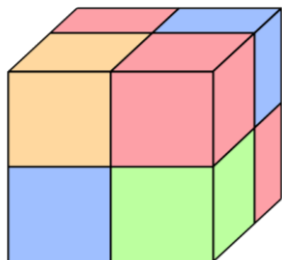
Voxel-based methods



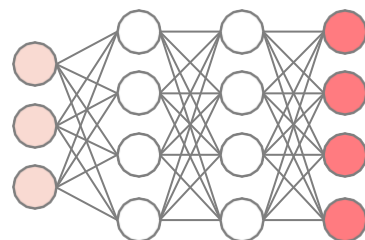
Trilinear Interpolation

Requirements

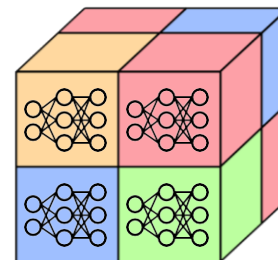
Scene
Representation



Voxelgrids



Implicit Function



Hybrid
Implicit/Explicit

...

Renderer

Volumetric

Sphere-Tracing
Volumetric

Volumetric

Pros

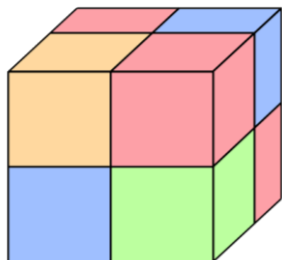
Fast rendering

Cons

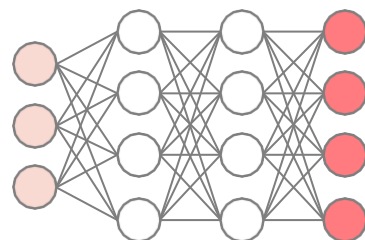
Memory $O(n^3)$
Limited spatial
resolution

Requirements

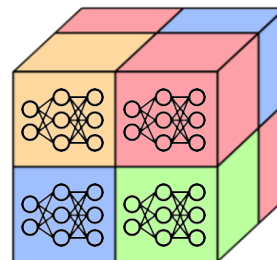
Scene
Representation



Voxelgrids



Implicit Function



Hybrid
Implicit/Explicit

...

Renderer

Volumetric

Sphere-Tracing
Volumetric

Volumetric

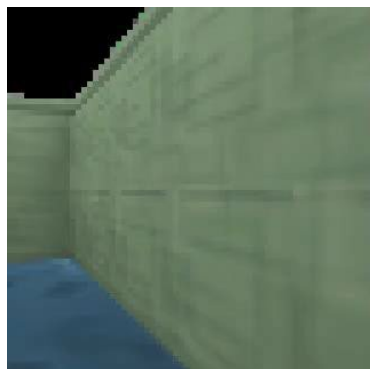
Pros

Fast rendering

Cons

Memory $O(n^3)$
Limited spatial
resolution

Neural Implicit Approaches



Scene Representation Networks
Generalizes across scenes
Sitzmann et al., NeurIPS 2019



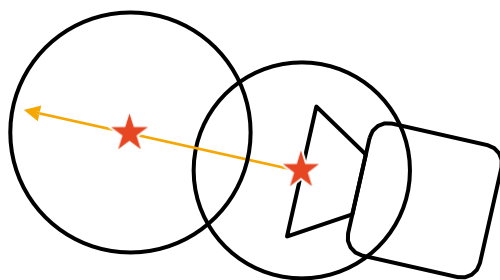
Differentiable Volumetric Rendering
Generalizes across scenes
Niemeyer et al., CVPR 2020



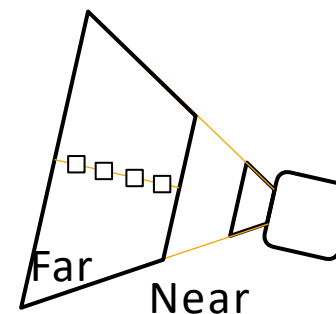
NeRF
Single-scene
Mildenhall et al., ECCV 2020



Implicit Differentiable Renderer
Single-scene
Yariv et al., NeurIPS 2020



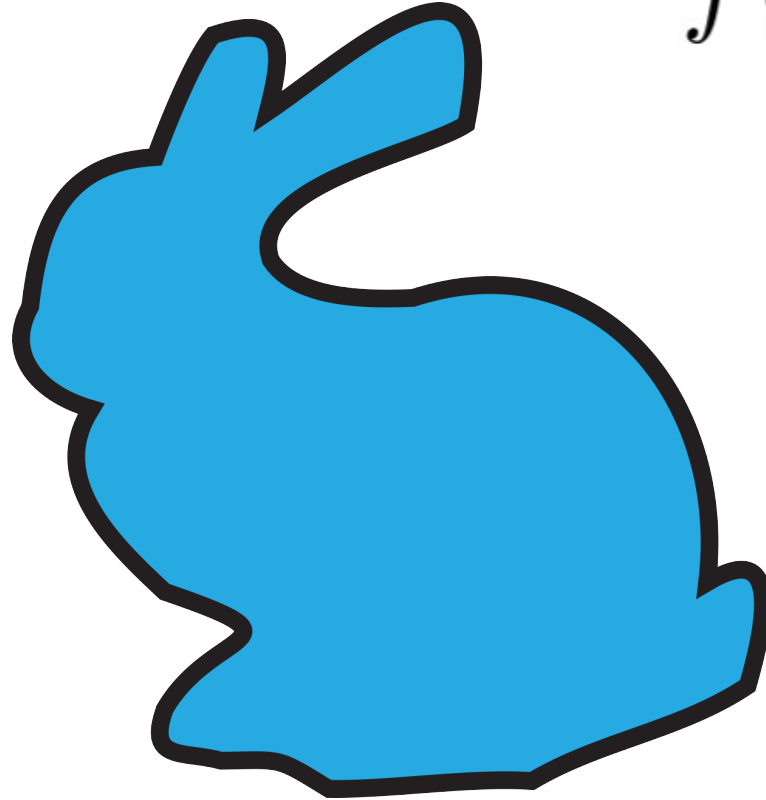
Sphere tracing



Volumetric

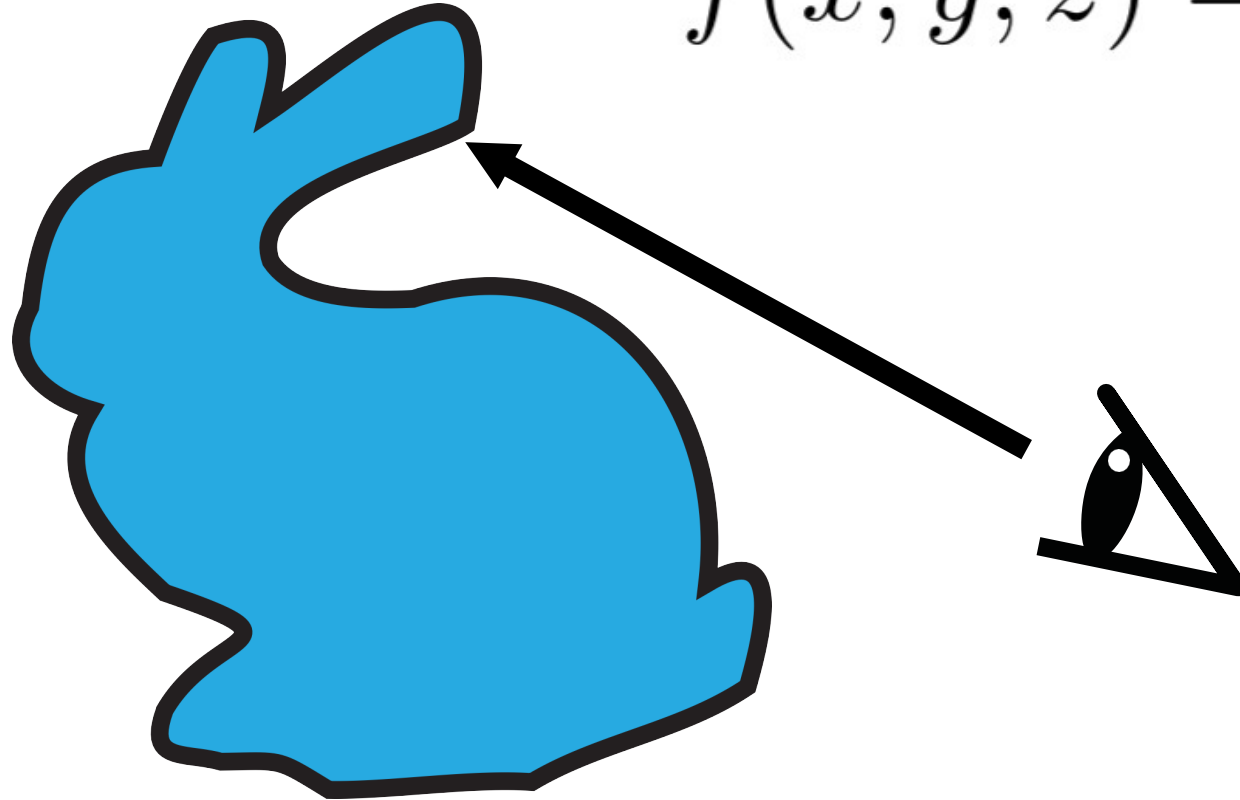
Sphere Tracing

$$f(x, y, z) = d$$



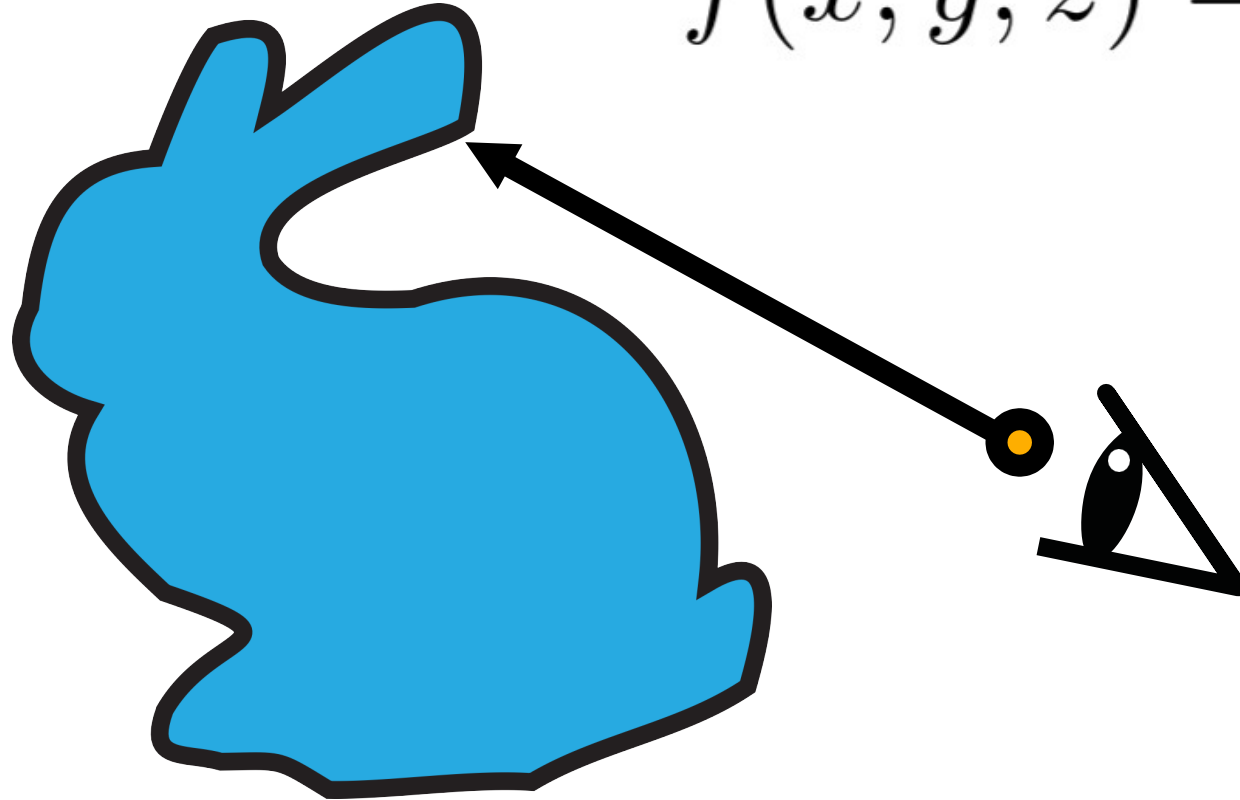
Sphere Tracing

$$f(x, y, z) = d$$



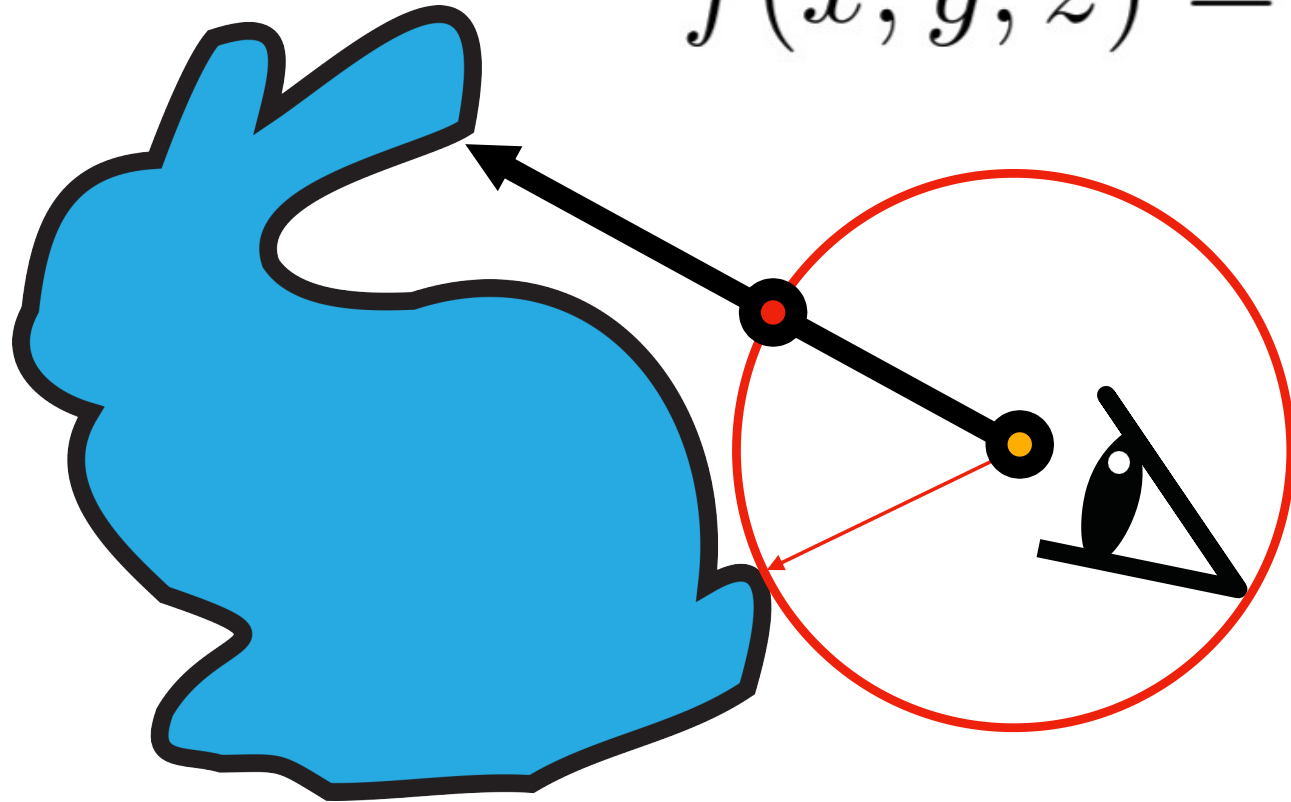
Sphere Tracing

$$f(x, y, z) = d$$

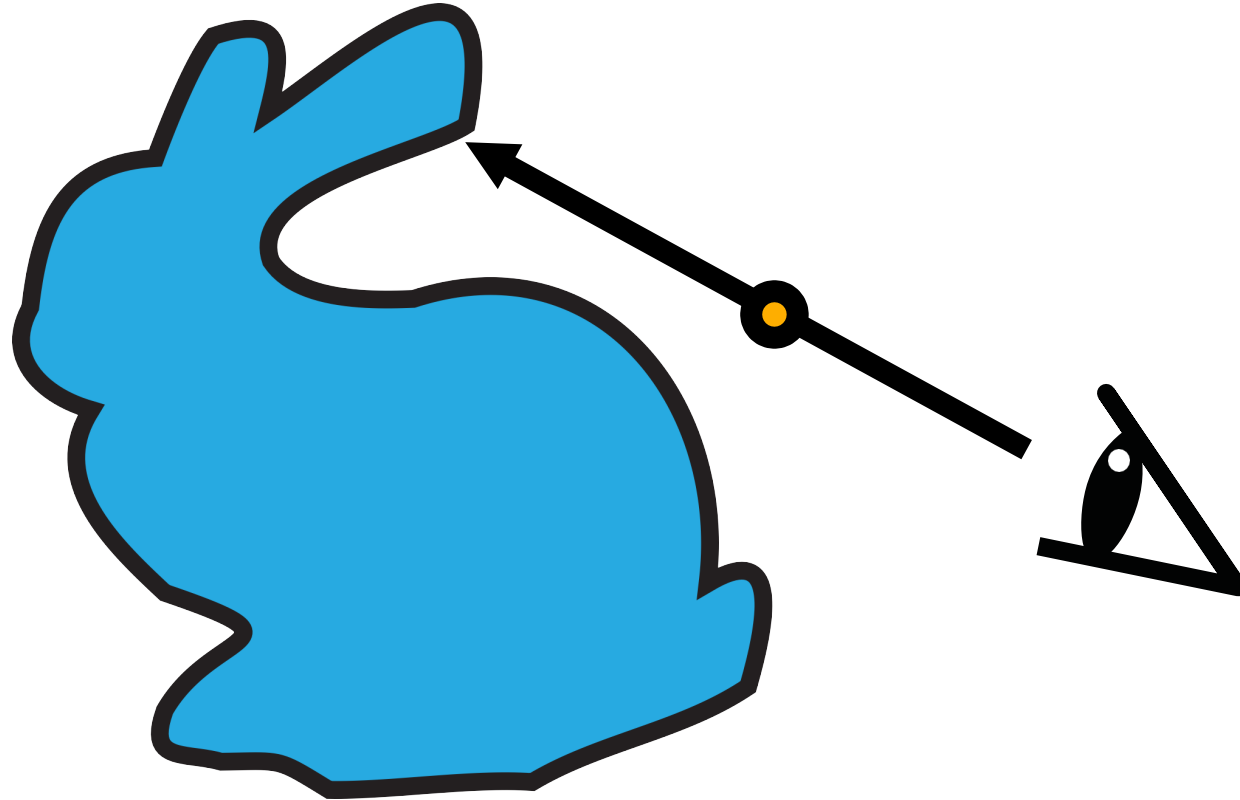


Sphere Tracing

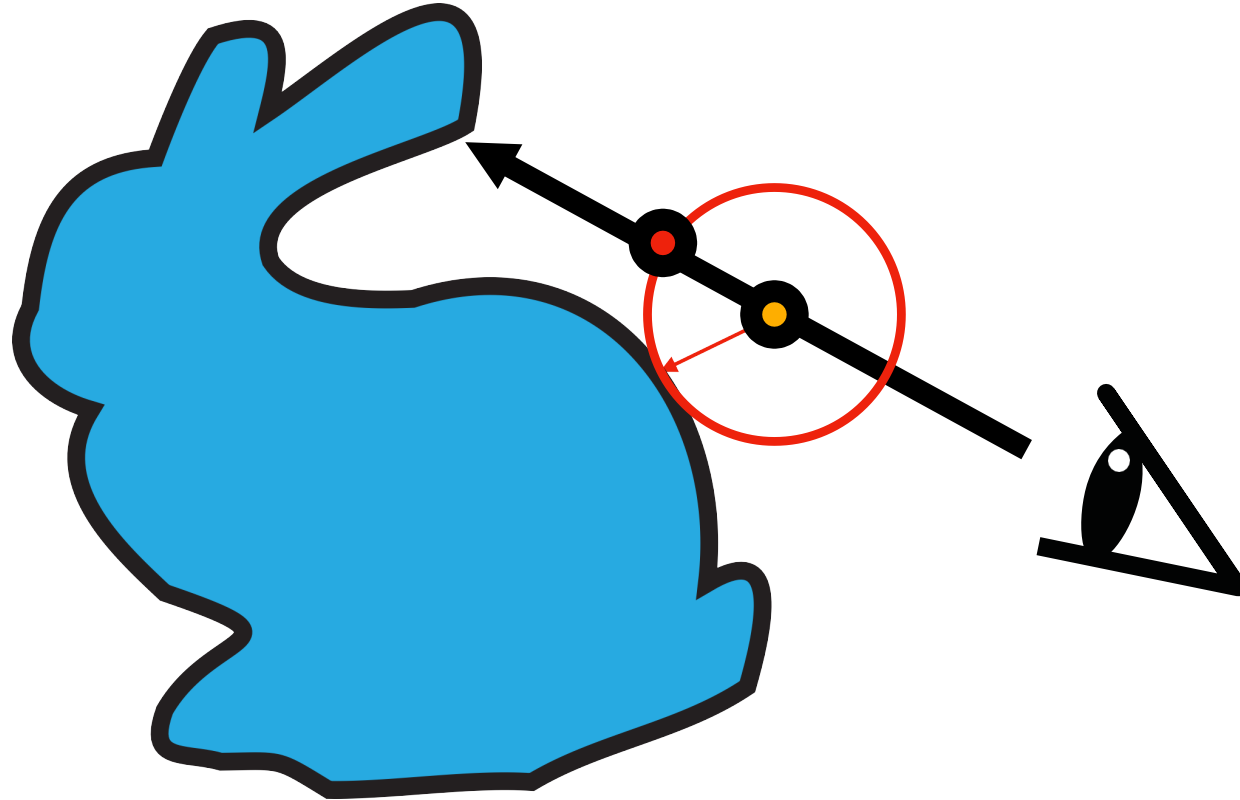
$$f(x, y, z) = d$$



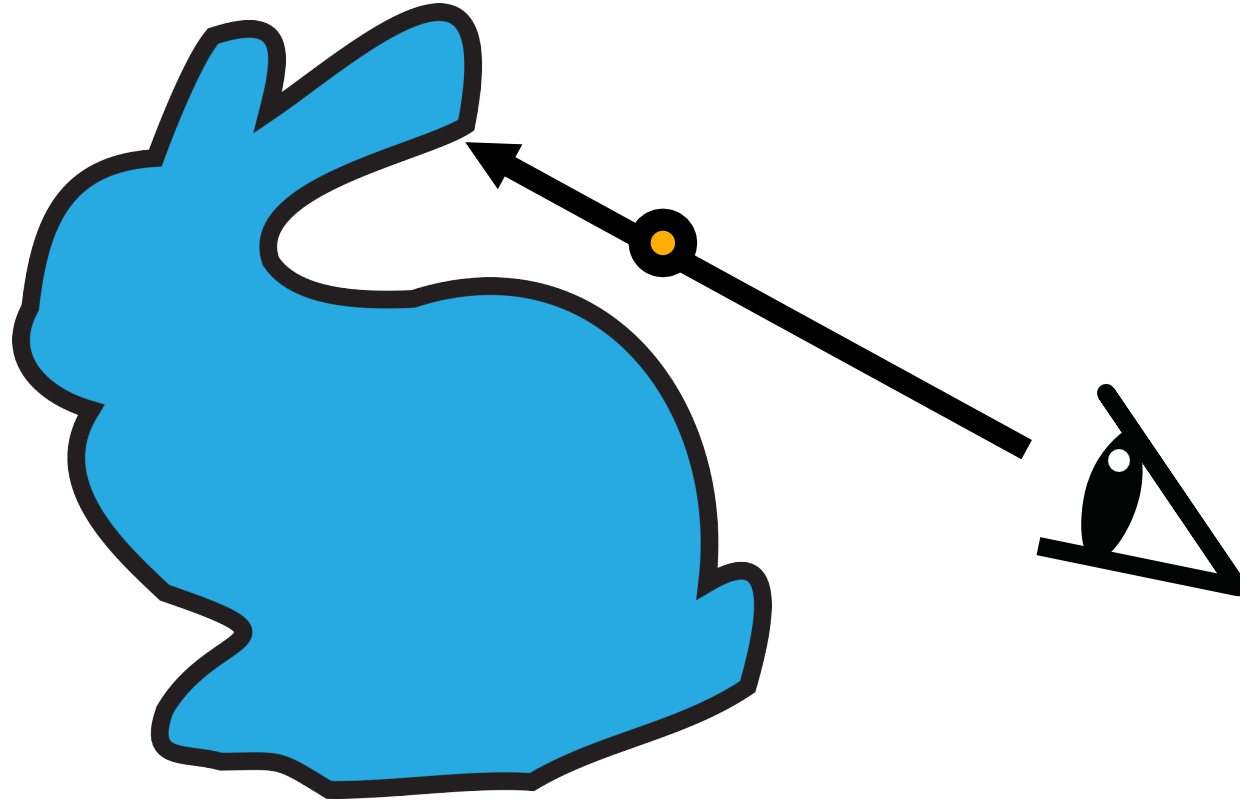
Sphere Tracing



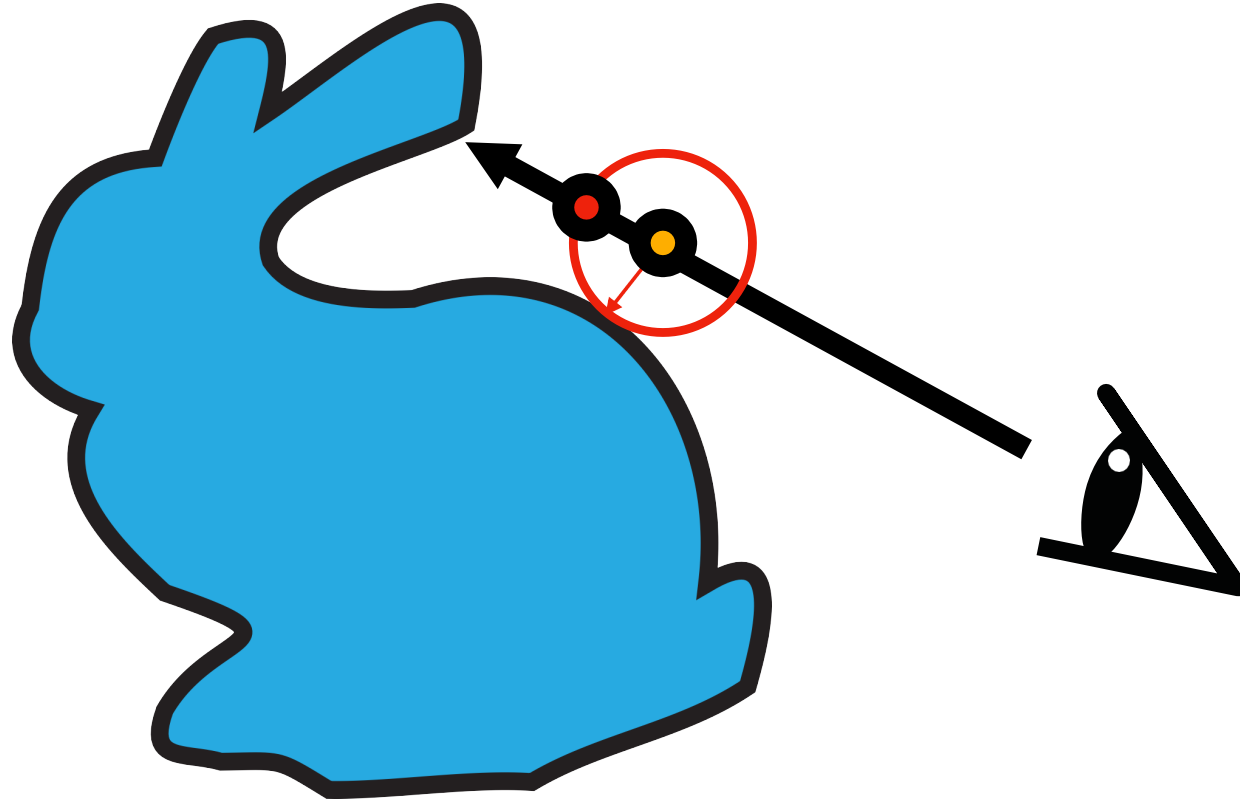
Sphere Tracing



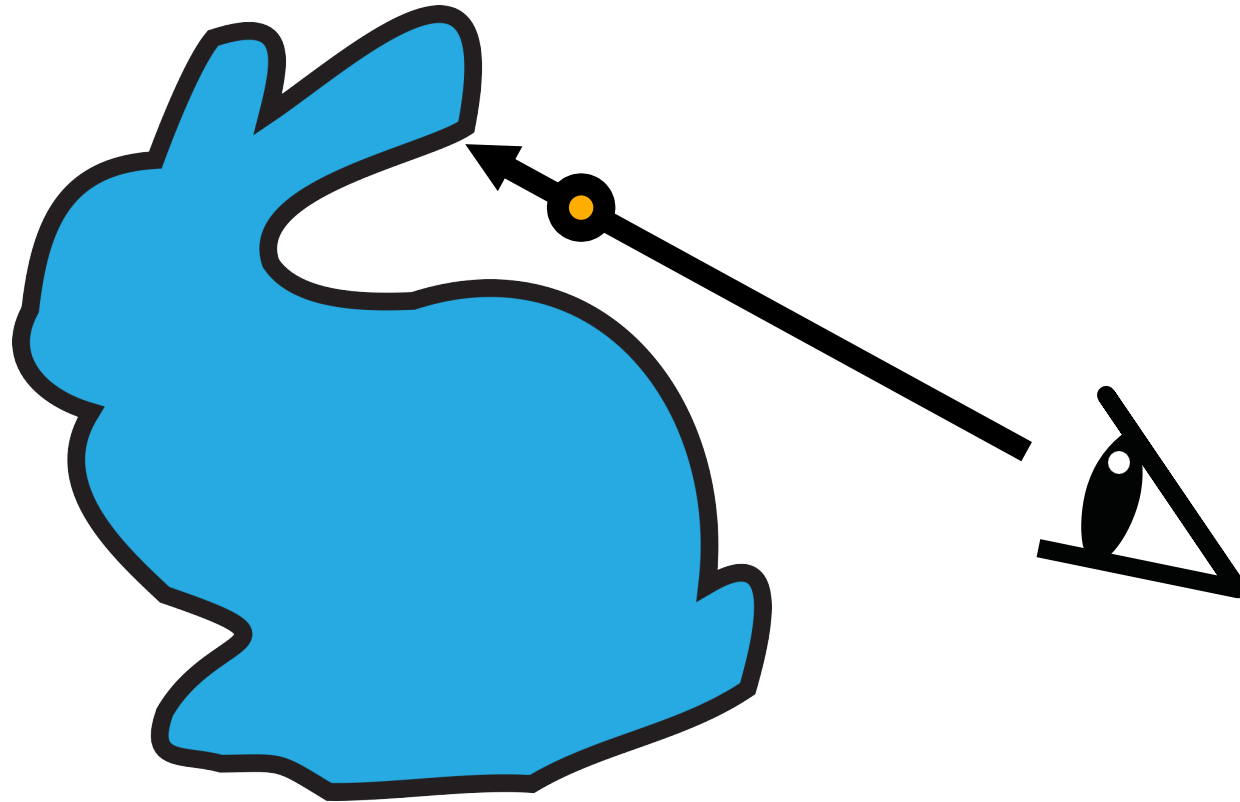
Sphere Tracing



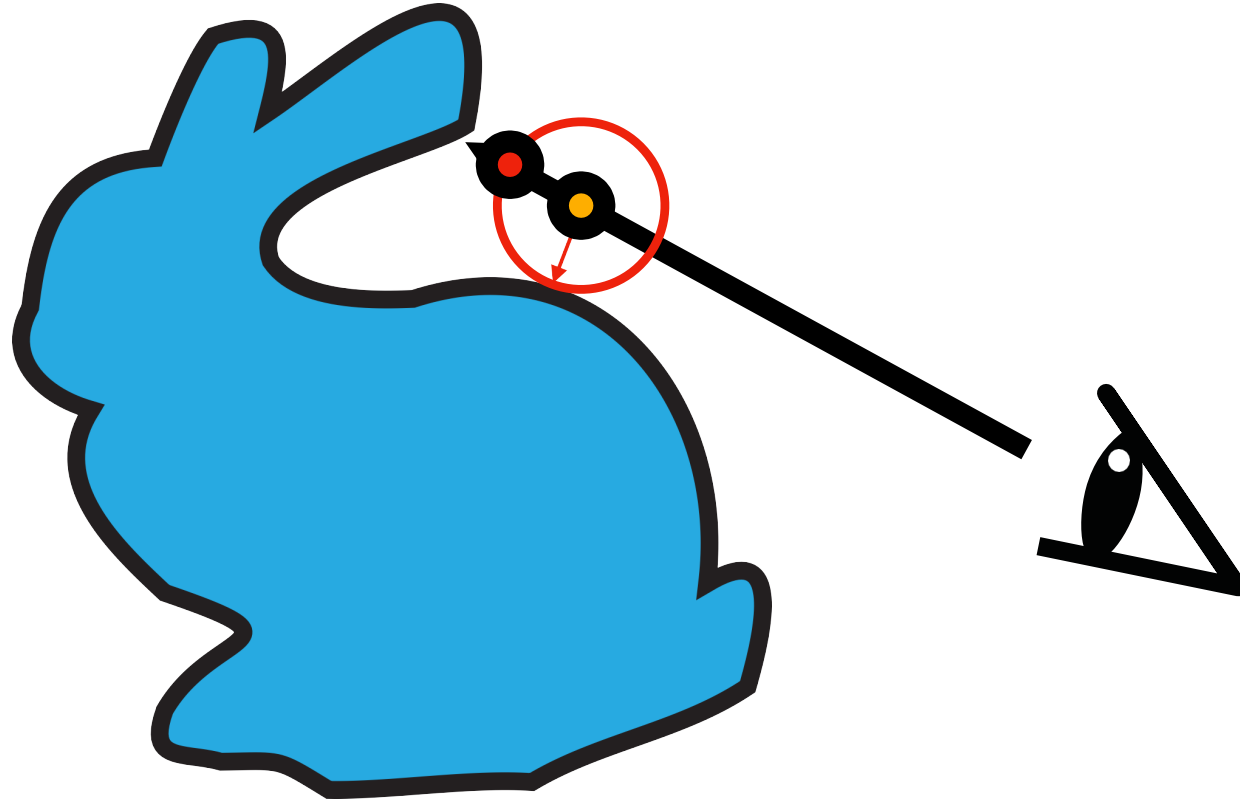
Sphere Tracing



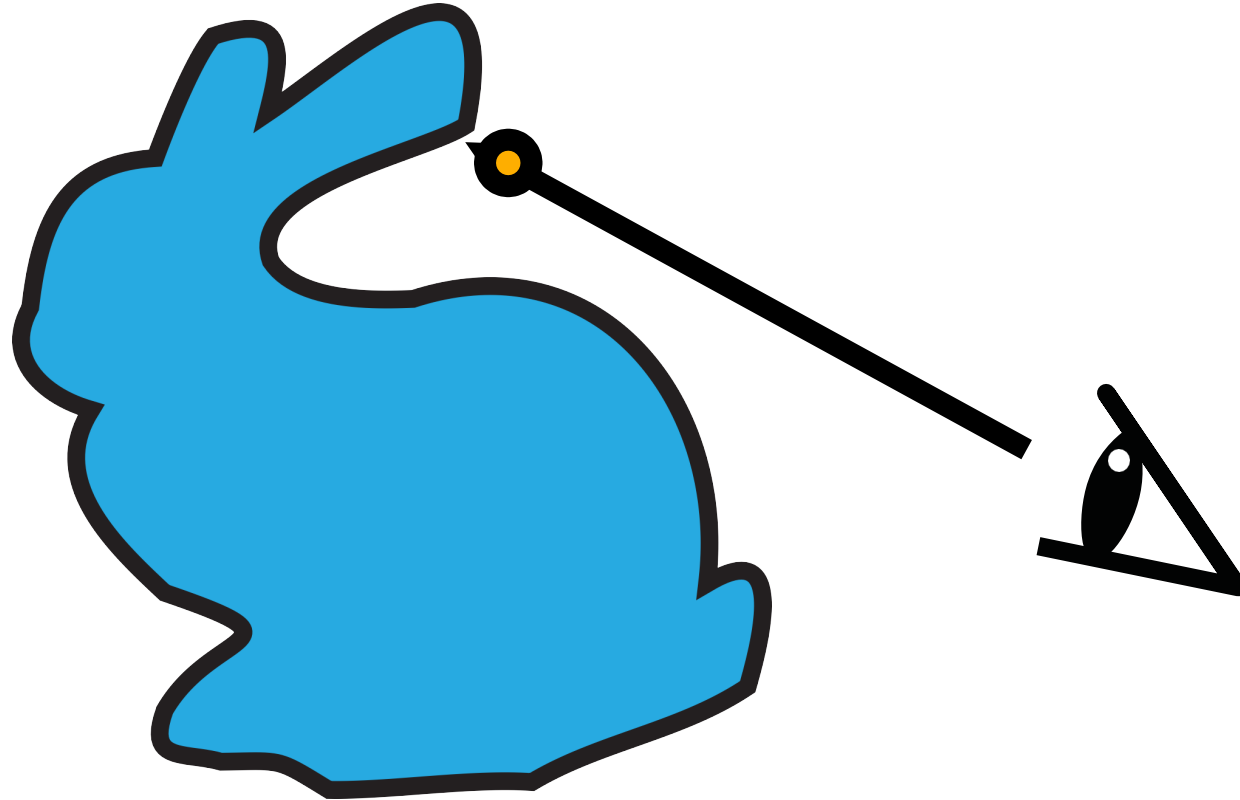
Sphere Tracing



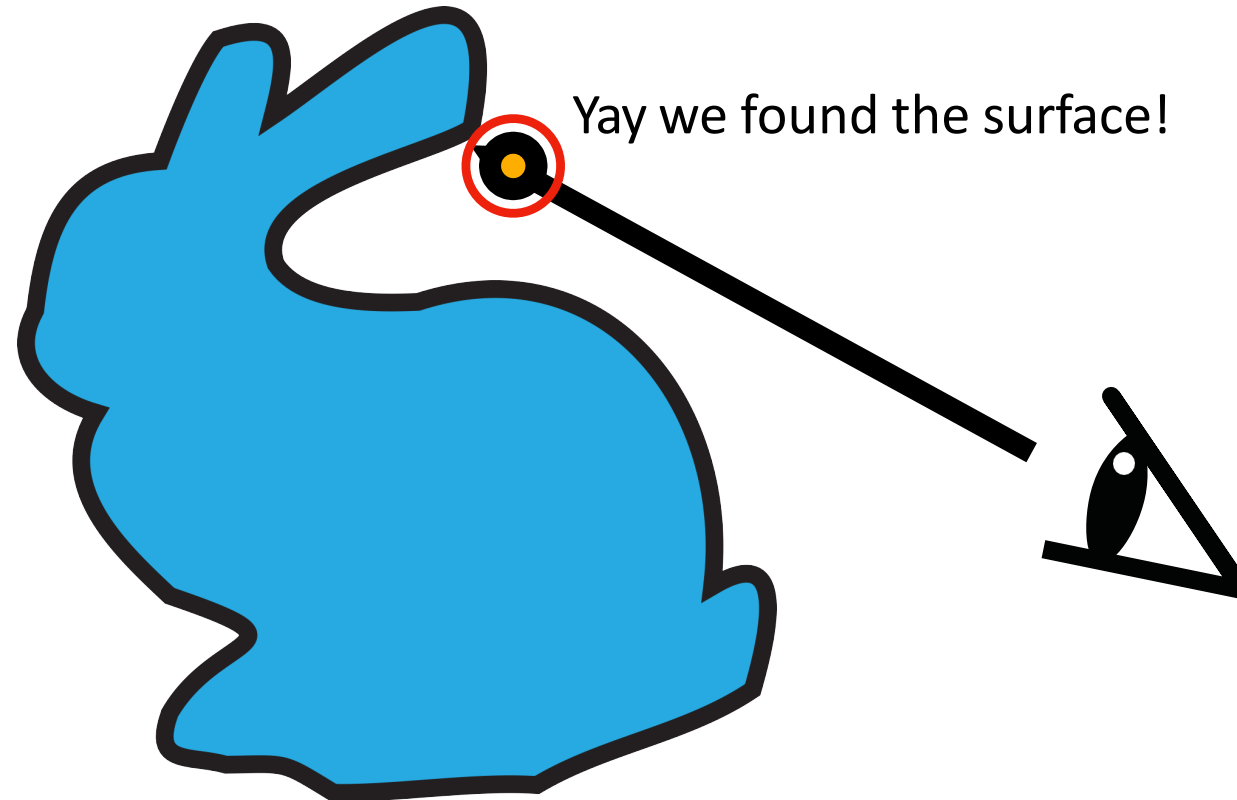
Sphere Tracing



Sphere Tracing

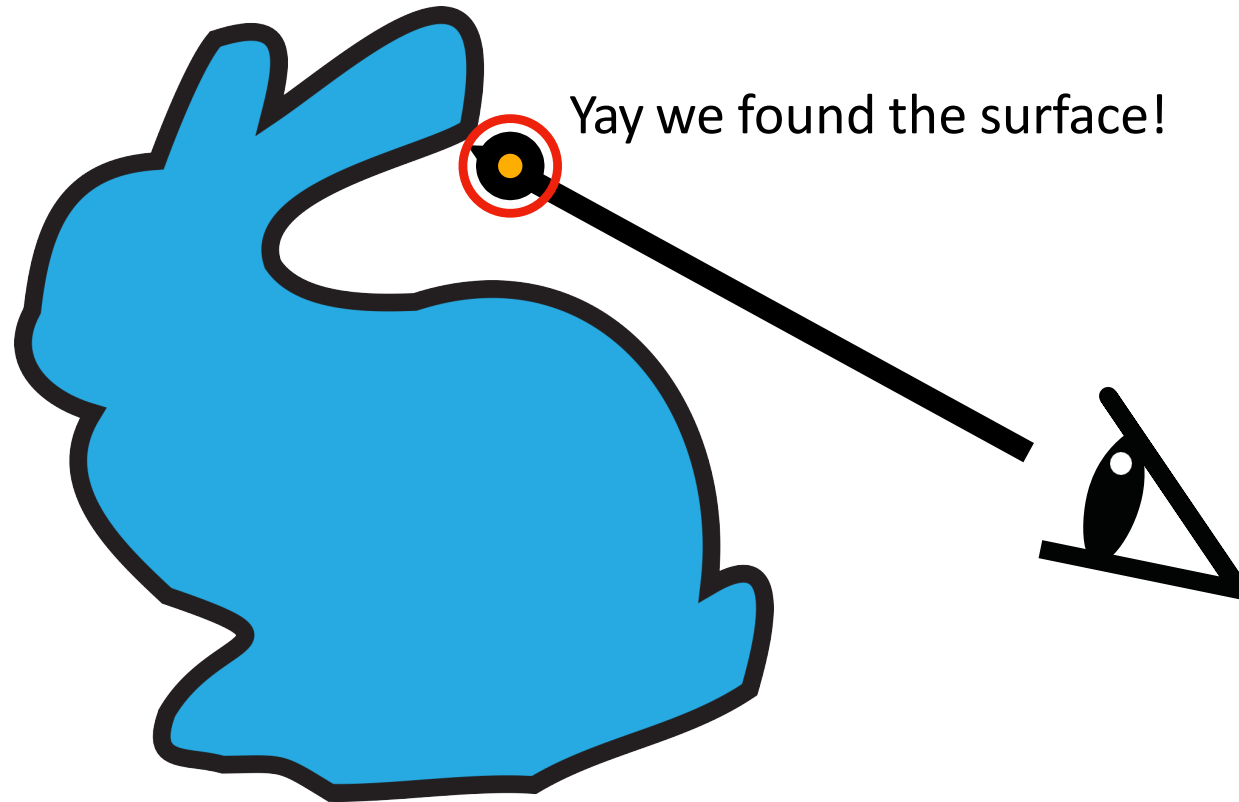


Sphere Tracing

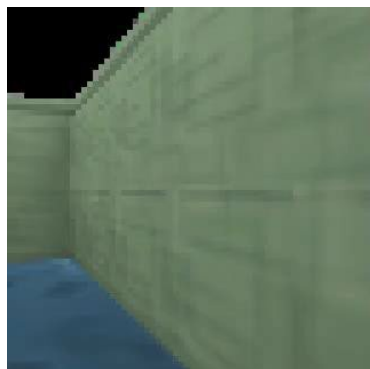


Sphere Tracing

$$f(x, y, z) = d$$



Neural Implicit Approaches



Scene Representation Networks
Generalizes across scenes
Sitzmann et al., NeurIPS 2019



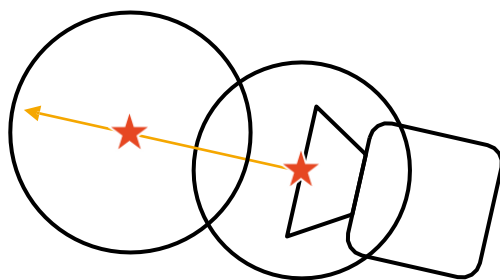
Differentiable Volumetric Rendering
Generalizes across scenes
Niemeyer et al., CVPR 2020



NeRF
Single-scene
Mildenhall et al., ECCV 2020

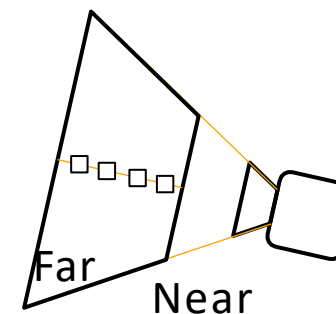


Implicit Differentiable Renderer
Single-scene
Yariv et al., NeurIPS 2020



Sphere tracing

- Faster
- Fewer network evaluations
- Convergence more difficult

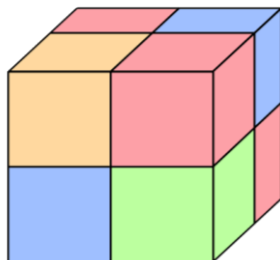


Volumetric

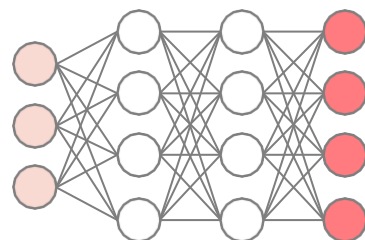
- Higher Quality
- Easy convergence
- Very expensive

Requirements

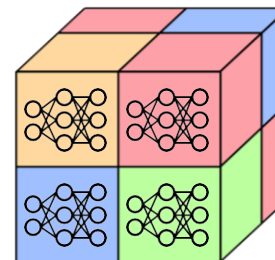
Scene
Representation



Voxelgrids



Implicit Function



Hybrid
Implicit/Explicit

...

Renderer

Volumetric

Sphere-Tracing
Volumetric

Volumetric

Pros

Fast rendering

High quality
Compact
Supports *global* priors

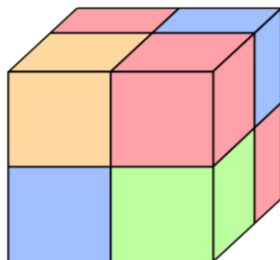
Cons

Memory $O(n^3)$
Limited spatial
resolution

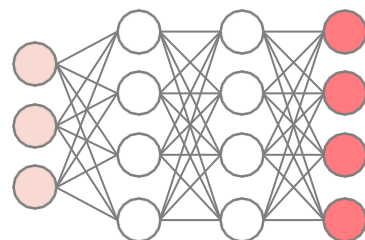
Extremely expensive,
slow rendering

Requirements

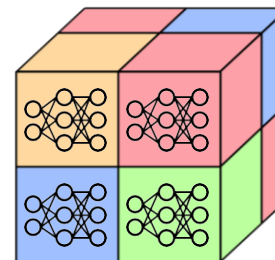
Scene
Representation



Voxelgrids



Implicit Function



Hybrid
Implicit/Explicit

...

Renderer

Volumetric

Sphere-Tracing
Volumetric

Volumetric

Pros

Fast rendering

High quality
Compact
Supports *global* priors

Cons

Memory $O(n^3)$
Limited spatial
resolution

Extremely expensive,
slow rendering

Hybrid Implicit / Explicit

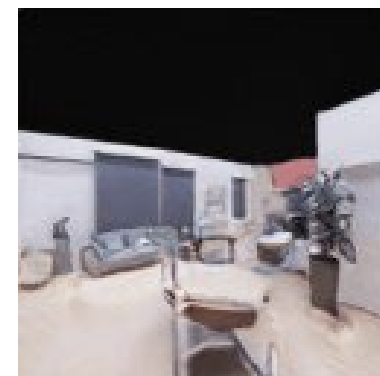
Wine Holder



Neural Sparse Voxel Fields,
Liu et. al., NeurIPS 2020



PiFU, Saito et al., ICCV 2019
GRF, Trevithick et al., arXiv 2020
pixelNeRF, Yu et. al., CVPR 2021
MVSNeRF, Chen et al., arXiv 2021
[Learn *local* \(image patch-based\) priors](#)



Unconstrained Scene Generation with
Locally Conditioned Radiance Fields,
DeVries et al., arXiv 2021

Neural Sparse Voxel Fields (NSVF)

- Avoid sampling points in empty space as much as possible.

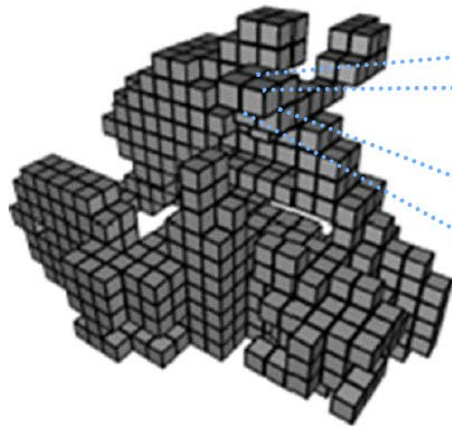


Illustration of Sparse Voxels

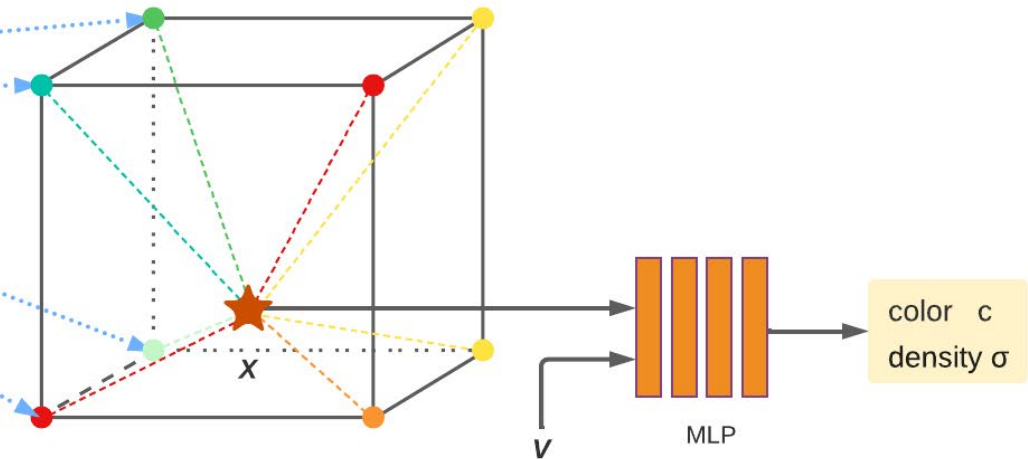
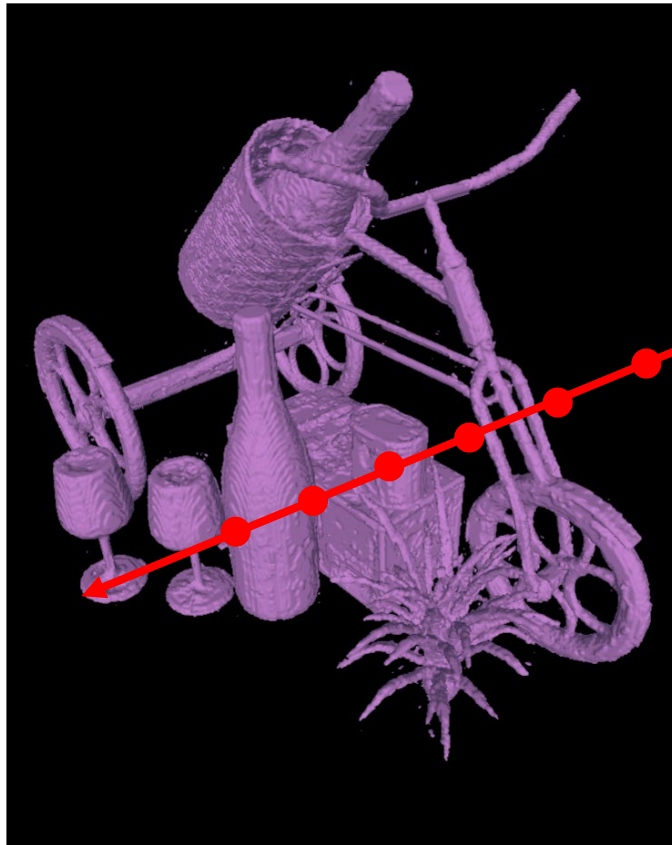


Illustration of a voxel-bounded neural field

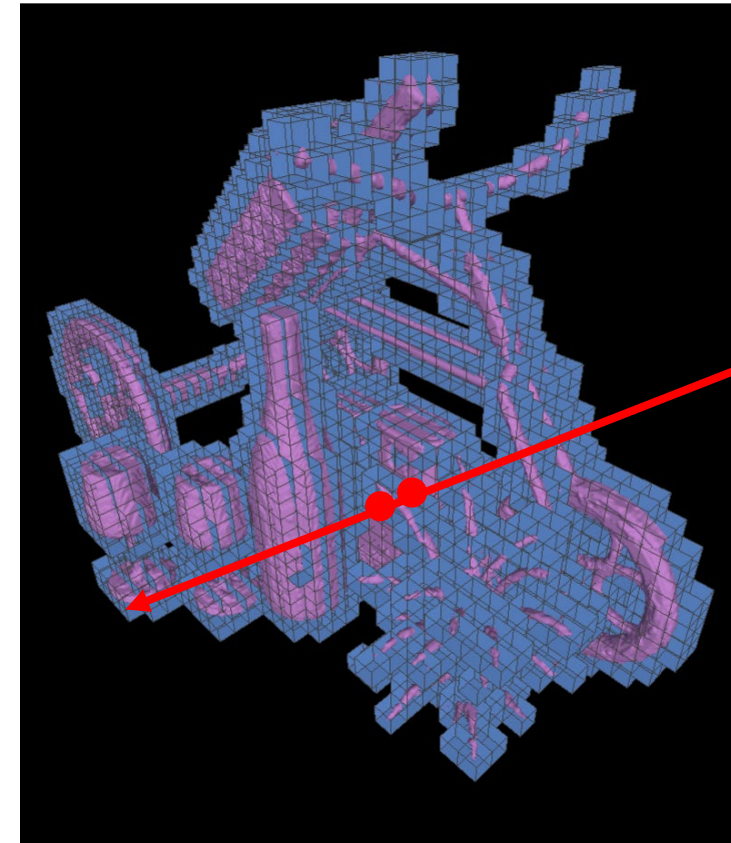
Neural Sparse Voxel Fields, Liu et al. 2020

Neural Sparse Voxel Fields (NSVF)

- Avoid sampling points in empty space as much as possible.



Sample in the whole space



Only sample inside the sparse-voxels

Comparison



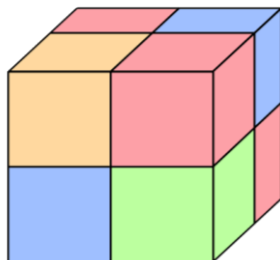
NeRF (Mildenhall et al. 2020)
(Rendering speed: 100 s/frame)



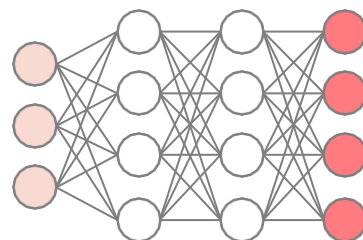
Ours (NSVF)
(Rendering speed: 2.62 s/frame)

Requirements

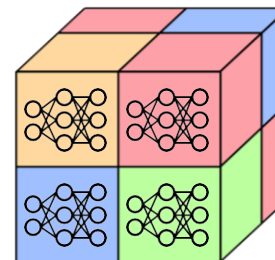
Scene
Representation



Voxelgrids



Implicit Function



Hybrid
Implicit/Explicit

...

Renderer

Volumetric

Sphere-Tracing
Volumetric

Volumetric

Pros

Fast rendering

High quality
Compact
Admits *global* priors

Significant Speedup
Supports *local* priors

Cons

Memory $O(n^3)$
Limited spatial
resolution

Extremely expensive,
slow rendering

No compact
representation No
global priors

Course Link:

<https://neural-representation-2026.github.io/index.html>



TAs



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Tentative Syllabus

	Presentation	Paper Presenting Schedule
1	Aug 25	Introduction
2	Aug 27	Introduction 2
3	Sep 1 (Tue)	Presentation Round 1 Start ->
4	Sep 3	
5	Sep 8 (Tue)	
6	Sep 10	
7	Sep 15 (Tue)	
8	Sep 17	
9	Sep 22 (Tue)	
10	Sep 24	
11	Sept 29 (Tue)	
	Oct 1	No Class. Fall term break.
12	Oct 6 (Tue)	Presentation Round 1 End <-

13	Oct 8	Guest Talk 1
14	Oct 13 (Tue)	Guest Talk 2
15	Oct 15	Guest Talk 3
16	Oct 20 (Tue)	Presentation Round 2 Start ->
17	Oct 22	
18	Oct 27 (Tue)	
19	Oct 29	
20	Nov 3 (Tue)	
21	Nov 5	
	Nov 10 (Tue)	No Class
	Nov 12	No Class
22	Nov 17 (Tue)	
23	Nov 19	
24	Nov 24 (Tue)	
	Nov 26	No Class
25	Dec 1 (Tue)	Presentation Round 2 End <-
26	Dec 3	Favorite Paper + New Ideas

Next Class

1. Present some pioneering works in this field, e.g., NeRF, SRN, Neural Volumes, Gaussian Splatting, NeuS, ...
2. Fundamentals of Classical 3D Representations and Rendering in Computer Graphics

Sep 1 (Tue)	W02-A	2	Neural Scene Representations & Learned Rendering	Classic bundle	Classic-centered	Core 3DV	Neural fields: MLP → hash grid → explicit grid Instant Neural Graphics Primitives with a Multiresolution Hash Encoding	ACM TOG / SIGGRAPH 2022
	W02-A	2	Neural Scene Representations & Learned Rendering	Classic bundle	Classic-centered	Core 3DV	Neural fields: MLP → hash grid → explicit grid Plenoxels: Radiance Fields without Neural Networks	CVPR 2022 Oral
Sep 3	W02-B	2	Neural Scene Representations & Learned Rendering	Single	Recent (2025-26)	Core 3DV	Minimal-inductive-bias view synthesis LVSM: A Large View Synthesis Model with Minimal 3D Inductive Bias	ICLR 2025 Oral / CVPR 2026 Highlight
	W02-B	2	Neural Scene Representations & Learned Rendering	Single	Recent (2025-26)	Core 3DV	Learned rendering with global illumination RenderFormer: Transformer-based Neural Rendering of Triangle Meshes with Global Illumination. RenderFormer++: Scalable and Physics-Informed Feed-Forward Neural Rendering	SIGGRAPH Asia 2025
Sep 8 (Tue)	W03-A	3	Implicit, Explicit & Structured Geometry	Classic bundle	Classic-centered	Core 3DV	SDFs and principled surface rendering Volume Rendering of Neural Implicit Surfaces	NeurIPS 2021 (VoISDF)
	W03-A	3	Implicit, Explicit & Structured Geometry	Single	Recent (2025-26)	Core 3DV	Structured Geometry Unifying Real-Time Differentiable Ray Tracing and Rasterization	Arxiv 2026
Sep 10	W03-B	3	Implicit, Explicit & Structured Geometry	Single	Recent (2025-26)	Core 3DV	Explicit Geometry Explicit mesh optimization with topology adaptation	CVPR 2026
	W03-B	3	Implicit, Explicit & Structured Geometry	Single	Recent (2025-26)	Core 3DV	Explicit Geometry Triangle splatting for real-time radiance field rendering MeshSplatting: Differentiable Rendering with Opaque Meshes	CVPR 2026
Sep 15 (Tue)	W04-A	4	Visual Geometry Foundation Models	Mixed bundle	Mixed lineage	Core 3DV	Point maps → general visual geometry transformer Lead — VGGT: Visual Geometry Grounded Transformer Skim — DUST3R: Geometric 3D Vision Made Easy. differt	CVPR 2025
	W04-A	4	Visual Geometry Foundation Models	Single	Recent (2025-26)	Core 3DV	Simple targets at foundation-model scale Depth Anything 3: Recovering the Visual Space from Any Views	ICLR 2026 Oral

Note: For a paper bundle, you only need to present one of the papers in the bundle according to their preference, but you are encouraged to discuss the connections between the papers in the bundle.

Before the seminar

- Read the papers of the week.
- Submit at least two questions for discussion before the seminar to a Google form (<https://docs.google.com/forms/d/e/1FAIpQLSfJ-UoJZQOpQS1ePEeezH8-c4zkkUDI1EhNzqXGJ9ExCPYoFg/viewform>). This is important – your contribution will be marked. The deadline for submitting questions is one hour before each class session (so Tuesday 12:45 PM and Thursday 12:45 PM).

During the seminar (Starting from Sept 1, two rounds)

- Overview (10 minutes)
 - The instructor or TAs give a brief introduction on the topic.
- 2x Presentations (each 25 minutes, 25 % of grade):
 - Two pre-assigned participants present the paper of their choice.
 - 5 minutes on motivation, background and related work.
 - 20 minutes of presentation of the paper.
- Discussion and Feedback (30 minutes, 25% of grade across weeks):
 - One participant is assigned at random at the beginning of the seminar to lead the discussion. Everyone leads the discussion at least once in the seminar series.
 - The discussion leader receives a digest of the submitted questions just before the seminar.
 - The discussion leader raises questions appropriately throughout the discussion, covers future work aspects, and finally provides a summary of the strengths and weaknesses of the techniques and of the discipline.
 - The students provide feedback to the presenting student on their presentation with respect to what has worked well, and what could be improved and how.

Grading Criteria

Form (30%) <i>To time? Verbal speed & clarity? Body posture? Engagement with audience?</i>	Moderation (30%) <i>Integrates questions well? Pushes forward discussion? Good summary? Strengths and weaknesses of paper?</i>	Practice Lecture (30%) <i>Listen attentively to the lecture? Engage in small coding exercises?</i>
Content (50%) <i>Structure/storyline? Main points? Paper connections? Valid conclusions?</i>	Questions (70%) <i>One question per paper (two questions per class) should be submitted at least one hour before the class during which the paper will be presented. (However, students are permitted to submit questions late (but before the discussion), up to two occurrences, without facing any penalties)</i>	Discussion on your favorite papers (35%) <i>Each person has 5 minutes to present their favorite paper. Is your presentation clear? Explain clearly why you chose it and what you like about it?</i>
Answers (20%) <i>Good answers to questions? Knowledgeable?</i>		Brainstorming (35%) <i>Actively participate in the discussion? Contribute your own ideas or opinions?</i>

2x Presentations
(50% of grade)

Discussion
(25% of grade)

Other Activity Participation
(25% of grade)

TODOs After this Class

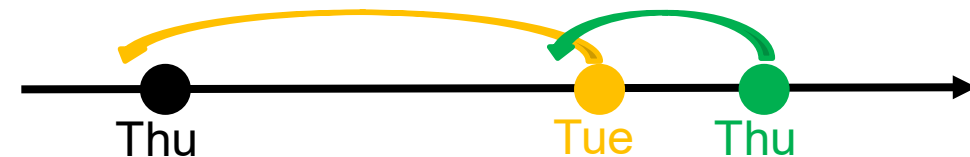
1. **Paper Selection and Registration:** [Important! Deadline: Aug 27]

Please select and register for the two papers you would like to present using the following Excel link:

<https://docs.google.com/spreadsheets/d/1qDT8KQOgKzVdRVaSnWr1UsH5S1WGbp75lvX92QaM124/edit?gid=333550507#gid=333550507>

2. **Presentation Preparation:**

- Please have your slides ready and upload them **at least one hour before the class before your presentation.**
- Please upload your slides to the Google folder: [link] (will send it to you later). This early deadline allows us to reschedule your presentation if an emergency arises.
- For example, if you're presenting on Tuesday, upload your slides by the previous Thursday at 12:45 PM. If presenting on Thursday, upload by Tuesday at 12:45 PM.



3. **Class Participation:**

- Before each class, please read the papers that will be discussed and submit two questions **at least one hour before the class** using the following link:

<https://docs.google.com/forms/d/e/1FAIpQLSfJ-UoJZQOpQS1ePEeezH8-c4zkkUDI1EhNzqXGJ9ExCPYoFg/viewform>

TODOs NOW

Find the first four speakers for the first two classes next week (Sep 1 and Sep 3).

Bonus: You can earn 3 bonus point for Tuesday's class and 2 bonus points for Thursday's class toward your final score!

Acknowledgments

- Advances in Neural Rendering
- Neural Fields in Visual Computing and Beyond
- awesome-NeRF: a curated list of awesome neural radiance fields papers
- MPII Summer Semester 2023: Computer Vision and Machine Learning for Computer Graphics

Any Questions?